

ChE 341 – Chemical Process Control –Spring 2026

Student Learning Outcome 1

Rubric for assessment problem on Exam #2 and Assignments #2, #6, #7

Assignment #6: Two Tanks in Series Level Control.

Assignment #7: High Order System, Skogestad Approximation Feedback Control

Assignment #2: Plant Wide Process Control

Exam #2: Miscellaneous Process Control Concepts

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Assignment #6 100 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<40 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (40-60 pts)	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (>60 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Assignment #7 100 pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<40 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (40-60 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (>60 pts)
4. Translates academic theory into engineering applications Assignment #2 100 pts	Does not appear to grasp the connection between theory and the problem (<40 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (40-60 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (>60 pts)
7. Solutions Creativity Alternatives Exam #2 100 pts	Demonstrates solutions implementing simple applications of one formula or equation with close analogies to class/lecture problems (<40 pts)	Demonstrates solution with integration of diverse concepts or derivation of useful relationships involving ideas covered in course concepts; however, no alternative solutions are generated (40-60 pts)	Demonstrates creative synthesis of solution and creates new alternatives by combining knowledge and information (>60 pts)

RESULTS BY PERFORMANCE INDICATOR				
Rubric #	1	2	4	7
Mapped SO	1	1	1	1
Average	2.72	2.89	2.89	2.89
Std Deviation	0.57	0.47	0.47	0.32
S	20	24	24	21
D	4	0	0	4
U	1	1	1	0
Checksum	25	25	25	25

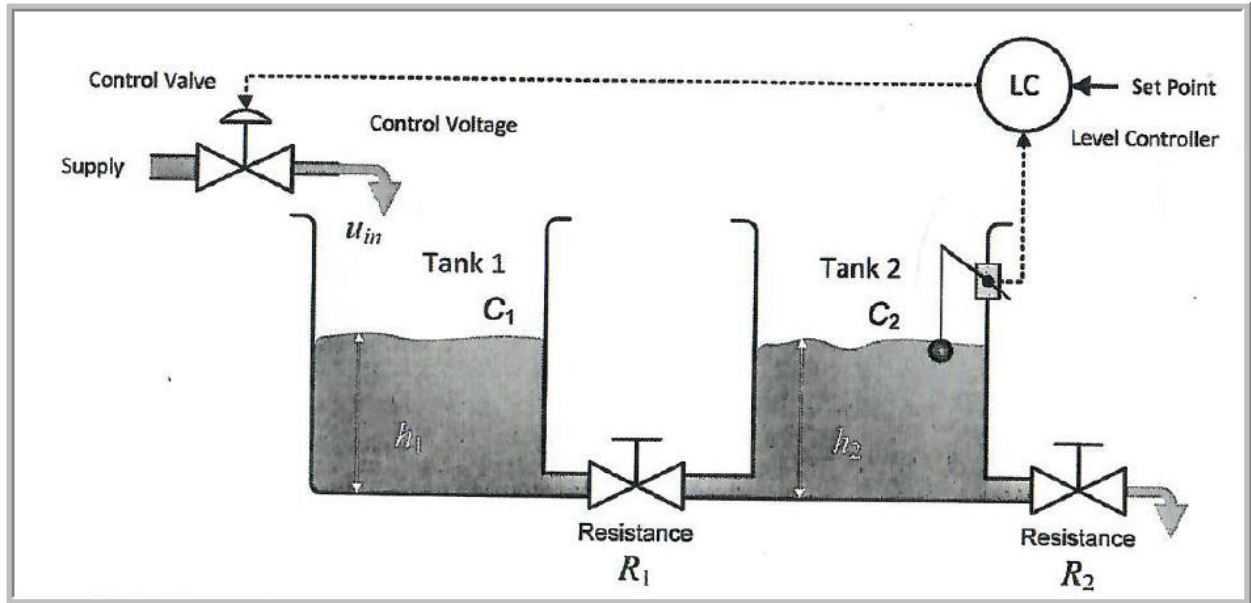
RESULTS BY STUDENT OUTCOME			
SO	1		
Average Score	2.85		
Standard Dev.	0.07		

Analysis of Results

Average scores were fairly uniform. No. 4 Performance Indicator has shown significant improvement from the last time, which is an indication on how students apply new ideas using their background obtained from my new textbook on process control. A large number of students were strong in solving engineering problems using theoretical process control concepts , and concepts of algebra, calculus, differential equations and/or linear algebra. Overall performance was satisfactory for all performance indicators.

Assignment #6: Two-Tank in Series Level Control System

In many chemical process applications, cascaded tank systems are used to attenuate disturbances and improve level regulation. Consider the two-tank system shown below, where liquid flows from Tank 1 into Tank 2.



A level transmitter is installed in Tank 2 and provides feedback to a PID controller, which manipulates the inlet flow rate to regulate the level in Tank 2.

The capacitance of the first and second tank is $C_1=C_2=1.0 \text{ m}^3/\text{m}$ and the outlet resistances of the first and second tank $R_1=0.5 \text{ m}/(\text{m}^3/\text{min})$ and $R_2=0.333 \text{ m}/(\text{m}^3/\text{min})$. The volumetric flow rate through the gate valves varies linearly with the pressure drop. Assume that the time delays in the control loop including the transportation lag between the first and second tank amount to around 35% of the dominant time constant of the system.

1. Derive the process and disturbance transfer functions assuming that the inlet flow acts as the manipulated variable (MV) and also as the disturbance variable (DV).
2. Design a PID controller with $GM=3$ and $PM=60$ degrees and perform a simulation for set point and disturbance change of unit size.
3. Compare your controller with:
 - a. Ziegler-Nichols Open Loop
 - b. Cohen-Coon
 - c. Ziegler-Nichols Closed Loop
 - d. Luyben-Tyreus
 - e. IMC-based tuning
 - f. Your own design

Present your results in a tabular form and compute the following metrics for both set point tracking and disturbance rejection of unit size.

1. Overshoot (if present)
2. Settling time
3. Gain Margin
4. Phase Margin
5. Maximum amount of time delay allowed before the onset of instability
6. Peak value M_s of the sensitivity function
7. Peak value M_t of the complementary sensitivity function

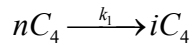
Include all the simulation graphs in your final report along with the Excel table shown below

	GM=3 PM=60 deg	Ziegler-Nichols Open Loop	Cohen-Coon	Ziegler-Nichols Closed Loop	Luyben-Tyreus	IMC	My Personal Controller	Selected Controller
Overshoot (if present)								
Settling time								
Gain Margin								
Phase Margin								
Maximum amount of time delay allowed before instability								
Peak Value of the sensitivity function								
Peak Value of the complimentary sensitivity function								
Controller Gain, K_c								
Controller Reset Time, τ_I								
Controller Derivative Time, τ_D								

Discuss the trade-offs observed between performance and robustness for the different controllers. Comment on which controller provides the best overall performance and justify your answer.

Assignment #2: Plant Wide Process Control.

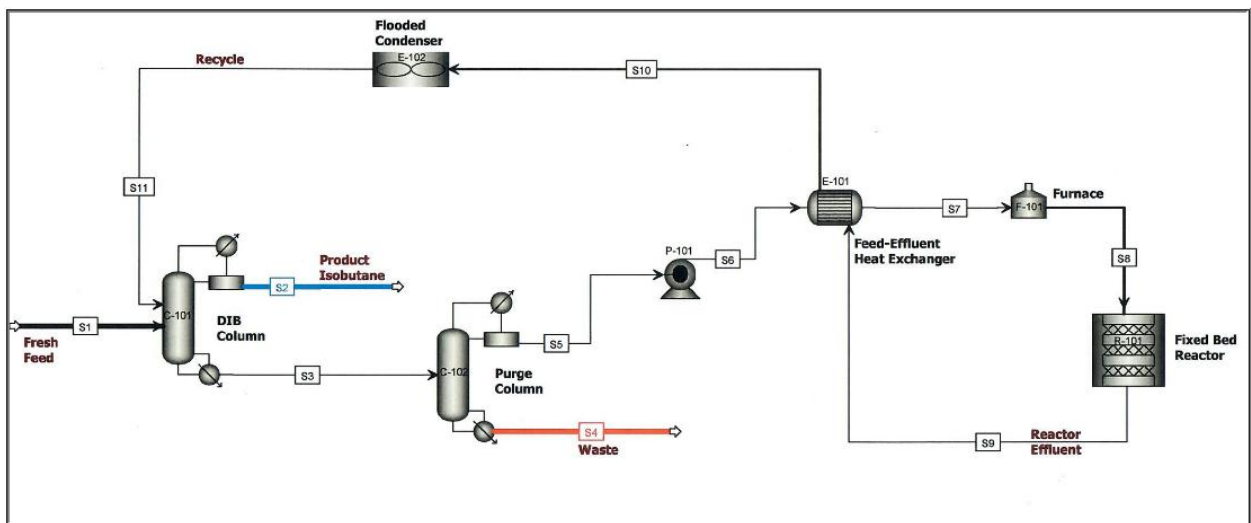
A process flowsheet diagram for the production of iso-butane via the isomerization of n-butane is shown in the Figure below: The isomerization reaction takes place in the fixed bed reactor with pressure drop equal to 15 psi. The kinetics of the reaction are shown below:



$$\mathcal{R}_1 = k_{10} \exp\left\{-\frac{E}{RT}\right\} C_{nC_4}$$

$$E = 30,000 \text{ Btu} / \text{lbmol}$$

$$k_{10} = 4 \times 10^8 \text{ 1/hr}$$



To better understand the process, open the attached ASPEN simulation and carefully study the operating conditions of all unit operations. The distillate product from the DeIsoButanizer column (DIB) has a specification of 2 mol% n-butane and is directed to a storage tank. The plant uses cooling water from near-by cooling tower designed with 75% efficiency with hot water inlet temperature of 120 °F and air ambient temperature of 90 °F with relative humidity 70%. A steam plant from a nearby facility supplies saturated steam (with quality 0.95) at three different levels of pressure: low pressure at 50 psig, medium pressure at 150 psig and high pressure at 600 psig. Natural gas from an existing pipe line is used for the fired heater that heats the feed to the reactor. The heat transfer coefficients in all the heat exchangers can be taken as the middle values of the ranges that are listed in the **CHE Calculator**.

Assume that the pressure drop in the heat exchanger is 10 psi both on the tube side and 5 psi on the shell side and the pressure drop in the fired heater is 60 psi. The pressure drop in the liquid line from the pump discharge to the inlet of the fired heater is approximately 8 psi.

This assignment has three parts:

Part A: Flowsheet Modifications

1. First, you must modify the simulation to obtain the product specifications and the diameters of the distillation columns. That will help you calculate the height of the liquid in the column sump. Some modifications will also be need to accommodate the additional information on pressure drop given in the problem description above.
2. Second you must create three utilities (cooling water, steam and natural gas) and attach them to the appropriate unit operations. Utility consumption will be used to calculate the steady-state values of the manipulated variables for the process control loops that you will identify in **Part B**.

Part B: Identification and Design of Process Control Loops

Following the procedure that I have outlined in the lectures, calculate the Control Degrees of Freedom for this process and design all process control loops using the additional information outlined below.

Data and Assumptions

1. Both refluxes in the **DIB** column and the **Purge** column are on flow control
2. Residence time in the reflux accumulation drums: 4 **min**
3. Residence time in the distillation column sumps: 2 **min**
4. The waste stream is also on flow control due to environmental restrictions
5. Neither the fresh feed nor the product flow rates are fixed by other plant considerations
6. Heating value of natural gas used in the fired heater: 50.0 **MJ/kg**
7. The heat capacity of water can be taken equal to 4.18 **kJ/(kg C)**
8. Temperature rise in the cooling water is limited to 11 °C

Process Variables Units for Reporting

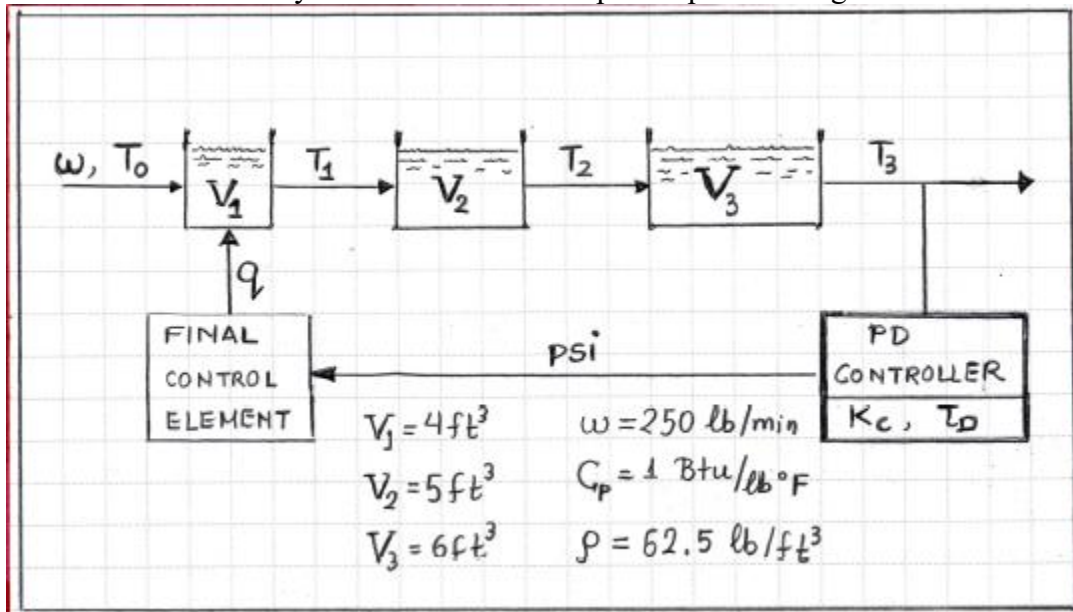
1. Liquid flow rates must be given in **gpm**
2. Gas flow rates in **lb/hr**
3. Temperature in **deg F**
4. Pressure in **psia**
5. Height in **ft**

For each control loop you have identified write a summary similar to the template shown

Loop #	Controller Description	Controlled Variable: CV	Set Point Value: SP	Manipulated Variable: MV	Steady-State Normal Value:
1	DIB Column Reflux Flow	Reflux Flow		Reflux Flow	
2					
...					

Assignment #7: Feedback Control of a Thermal System

Consider the thermal system shown in the simplified process diagram below:



The thermal system is currently being controlled by a **Proportional Derivative (PD)**

controller with gain and rate equal to to: $K_C = 1500 \frac{\text{Btu/min}}{^\circ\text{F}}$ $\tau_D = 0.5 \text{ min}$ The

temperature T_o of the inlet stream is considered a disturbance.

1. Draw a block diagram of the control system with the transfer functions in each block. Each transfer function must only have the numerical values of its parameters.
2. From this block diagram, determine the closed loop transfer function relating the temperature in tank #3 to a change in the set point.
3. From this block diagram determine the closed loop transfer function relating the temperature in tank #3 to a change in the inlet temperature T_o .
4. Calculate the offset for a unit step change in the inlet temperature T_o .
5. Perform a simulation using the **CHE Calculator.xlsm** Excel tool to verify your theoretical answer in question #4.
6. You have been asked to eliminate the observed offset. Design a controller and calculate its parameters that would accomplish just that. Perform a simulation again, using the **CHE Calculator.xlsm** Excel tool to verify that the observed offset has been eliminated.
7. Calculate the settling time and present it along with the controller parameters in a Table included in a separate page in your final report. Your final report should also include the graphical results for questions #5 and #6.

Assignment #7 Solution: Feedback Control of a Thermal System

- A. We begin by developing the process & disturbance transfer functions

Tank #1 Disturbance variable is the feed temperature T_0

Manipulate variable is the heat flow q

Process Variable is the effluent temperature

We now perform an energy balance around tank #1

$$\rho V_1 C_p \frac{dT_1}{dt} = w C_p (T_0 - T_1) + q \quad \text{Define } \frac{\rho V_1}{w} = \tau_1 \quad \frac{q}{w C_p} = \frac{q}{250} \Rightarrow$$

$$\tau_1 \frac{dT_1}{dt} = T_0 - T_1 + \frac{q}{250} \quad (\text{Eq. 1})$$

$$\text{At steady-state } \tau_1 \frac{dT_1^{ss}}{dt} = T_0^{ss} - T_1^{ss} + \frac{q^{ss}}{250} \quad (\text{Eq. 2})$$

Define the deviation variables:

Process Variable Deviation: $Y_1 = T_1 - T_1^{ss}$ Process Gain $K_p = \frac{1}{250}$

Manipulated Variable Deviation: $X = q - q^{ss}$

Disturbance Variable Deviation $D = T_0 - T_0^{ss}$

Subtracting (Eq. 2) from (Eq. 1) and using deviation variable

$$\tau_1 \frac{dY_1}{dt} + Y_1 = K_p X + D \quad \text{since } \tau_1 = \frac{\rho V_1}{w} = \frac{(62.5)(4)}{250} = 1 \text{ min}$$

$$\frac{dY_1}{dt} + Y_1 = K_p X + D \Rightarrow s Y_1(s) + Y_1(s) = K_p X(s) + D(s) \Rightarrow Y_1(s) = \frac{K_p X(s)}{s+1} + \frac{1}{s+1} D(s)$$

Therefore from the analysis of tank #1 we obtain its process transfer function G_{p1} and disturbance transfer function G_d where

$$G_{p1} = \frac{Y_1(s)}{X(s)} = \frac{0.004}{s+1} \quad G_d = \frac{Y_1(s)}{D(s)} = \frac{1}{s+1}$$

Tanks #2 and #3

Writing energy balances for Tanks #2 and #3 we get:

$$\rho V_2 C_p \frac{dT_2}{dt} = w C_p (T_1 - T_2) \quad \text{and} \quad \rho V_3 \frac{dT_3}{dt} = w C_p (T_2 - T_3)$$

Introducing again deviation variables $Y_2 = T_2 - T_2^{ss}$, $Y_3 = T_3 - T_3^{ss}$ we obtain

$$\frac{\rho V_2}{w} \frac{dY_2}{dt} = Y_1 - Y_2 \Rightarrow \tau_2 \frac{dY_2}{dt} + Y_2 = Y_1 \Rightarrow \frac{Y_2(s)}{Y_1(s)} = \frac{1}{\tau_2 s + 1}$$

$$\text{Since } \tau_2 = \frac{\rho V_2}{w} = \frac{(62.5)(5)}{250} \Rightarrow \tau_2 = 1.25 \text{ min}$$

$$\text{and therefore } G_{p2} = \frac{Y_2(s)}{Y_1(s)} = \frac{1}{1.25s + 1}$$

Following similar steps for tank #3 we obtain

$$\frac{Y_3(s)}{Y_2(s)} = \frac{1}{\tau_3 s + 1} \quad \text{and since } \tau_3 = \frac{\rho V_3}{w} = \frac{(62.5)(6)}{250} \Rightarrow \tau_3 = 1.5$$

$$\text{we get } G_{p3} = \frac{Y_3(s)}{Y_2(s)} = \frac{1}{1.5s + 1}$$

To summarize the overall process transfer function is $G_p = G_{p1} G_{p2} G_{p3} \Rightarrow$

$$G_p = \frac{Y_3(s)}{Q(s)} = \frac{K_p (=0.004)}{(1.5s+1)(1.25s+1)(s+1)}$$

Setting $Y_3(s) = PV$ (Process (controlled) Variable)

$Q(s) = MV$ (Manipulated Variable)

$T_o(s) = DV$ (Disturbance Variable)

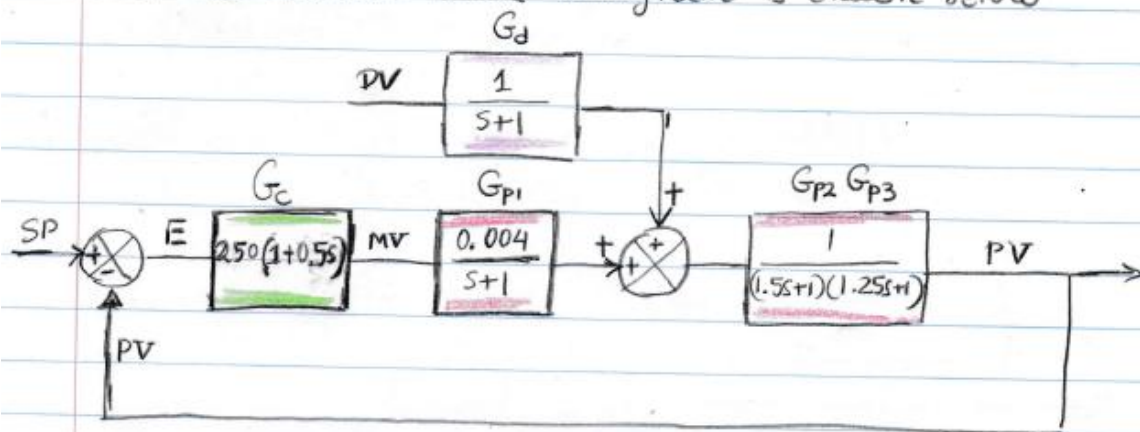
we get

$$G_p(s) = \frac{PV(s)}{MV(s)} = \frac{K_p (=0.004)}{(1.5s+1)(1.25s+1)(s+1)} \quad G_d(s) = \frac{PV}{DV} = \frac{1}{s+1}$$

The Controller Transfer function is

$$G_c = K_c (1 + T_D s) \Rightarrow G_c = 250 (1 + 0.5s)$$

and the Control Block Diagram is drawn below



B. Closed Loop Transfer function $\frac{PV}{SP}$ (Set $DV=0$)
Looking at the control block diagram we have

$$PV = (MV) \cdot G_{p1} G_{p2} G_{p3} = (MV) \cdot G_p$$

$$MV = (SP - PV) G_c$$

Combining the two equations

$$PV = (SP - PV) G_c G_p \Rightarrow \frac{PV}{MV} = \frac{G_c G_p}{1 + G_c G_p} \Rightarrow$$

$$\frac{PV}{SP} = \frac{250(1+0.5s) \cdot \frac{0.004}{(1.5s+1)(1.25s+1)(s+1)}}{1 + 250(1+0.5s) \cdot \frac{0.004}{(1.5s+1)(1.25s+1)(s+1)}} \Rightarrow$$

$$\frac{PV}{SP} = \frac{0.5s+1}{(1.5s+1)(1.25s+1)(s+1) + (0.5s+1)} \quad (\text{Eq. 1})$$

• Closed Loop Transfer function $\frac{PV}{DV}$ (Set $SP=0$)

$$PV = [(MV) G_{p1} + (DV) G_d] G_{p2} G_{p3}$$

$$MV = -(PV) G_c$$

Combining the two equations we obtain

$$PV = [-(PV) G_c G_{p1} G_{p2} G_{p3}] + (DV) G_d G_{p2} G_{p3} \Rightarrow$$

$$PV [1 + G_c G_{p1} G_{p2} G_{p3}] = DV G_d G_{p2} G_{p3} \Rightarrow$$

$$\frac{PV}{DV} = \frac{G_d G_{p2} G_{p3}}{1 + G_c G_p} \Rightarrow \frac{PV}{DV} = \frac{\frac{1}{(1.5s+1)(1.25s+1)(s+1)}}{(1.5s+1)(1.25s+1)(s+1) + (0.5s+1)} \Rightarrow$$

$$\frac{PV}{DV} = \frac{1}{(1.5s+1)(1.25s+1)(s+1) + (0.5s+1)}$$

D. From the closed loop transfer function we get for a unit step in the disturbance

$$PV(s) = \frac{1}{s} \frac{1}{(1.5s+1)(1.25s+1)(s+1)+(0.5s+1)}$$

Applying the final value theorem we have

$$PV(t \rightarrow \infty) = \lim_{s \rightarrow 0} s \cdot PV(s) \Rightarrow \boxed{PV(t \rightarrow \infty) = 0.5}$$

E. To use the CHE341 Calculator we have the following Data

Disturbance: Gain = 1

Lag Time = 1

Dead Time = 0

Disturbance active on CONTROL VALVE OUTPUT

Control Valve: Gain = 0.004

Lag Time = 1

Controller: Gain = 250

Integral Time = 0

Derivative Time = 0.5

PROCESS:

Transfer function $G_{p2} G_{p3} = \frac{1}{(1.5s+1)(1.25s+1)}$

Applying the Skogestad approximation

$$\tau^* = 1.5 + \frac{1.25}{2} = 2.125 \quad \text{and} \quad \theta^* = 0.625 \Rightarrow G_{p2} G_{p3} \approx \frac{e^{-0.625s}}{2.125s+1}$$

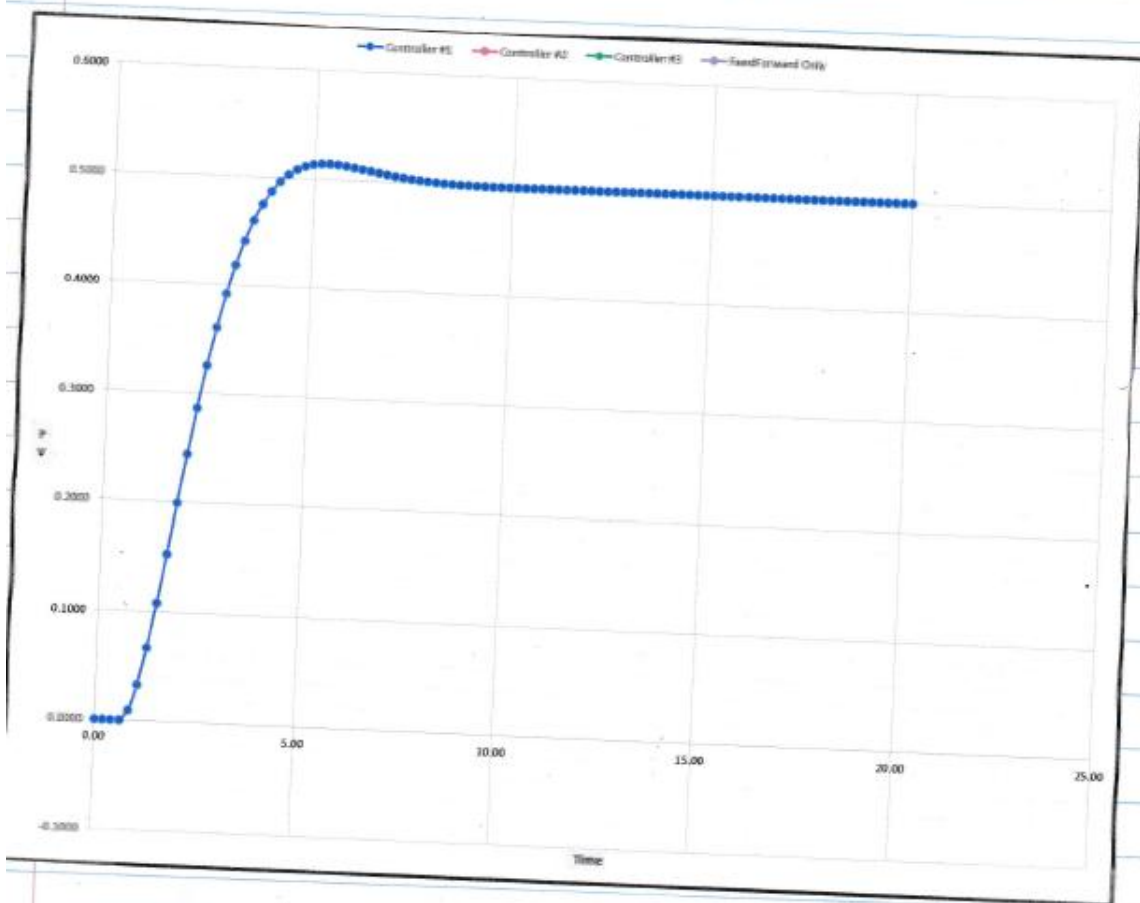
Therefore for the process in the ChE Calculator we have

Process: Gain = 1

Lag Time = 2.125

Dead time = 0.625

When we enter all data in the ChE341 Calculator we obtain the graph shown below that verifies our answer in Part D.



F. To design a controller that eliminates the offset we first must consider the overall process transfer function $G_p = \frac{0.004}{(1.5s+1)(1.25s+1)(s+1)}$

Now we use Skogestad's rule to develop an FOPT model

$$\tau^* = 1.5 + 1.25/2 = 2.125$$

$$\theta^* = 1.25/2 + 1 = 1.625$$

We now use these values to design a PI controller using the IMC tuning method with $T_c = \theta^*$ we obtain $K_c = 163.4615$

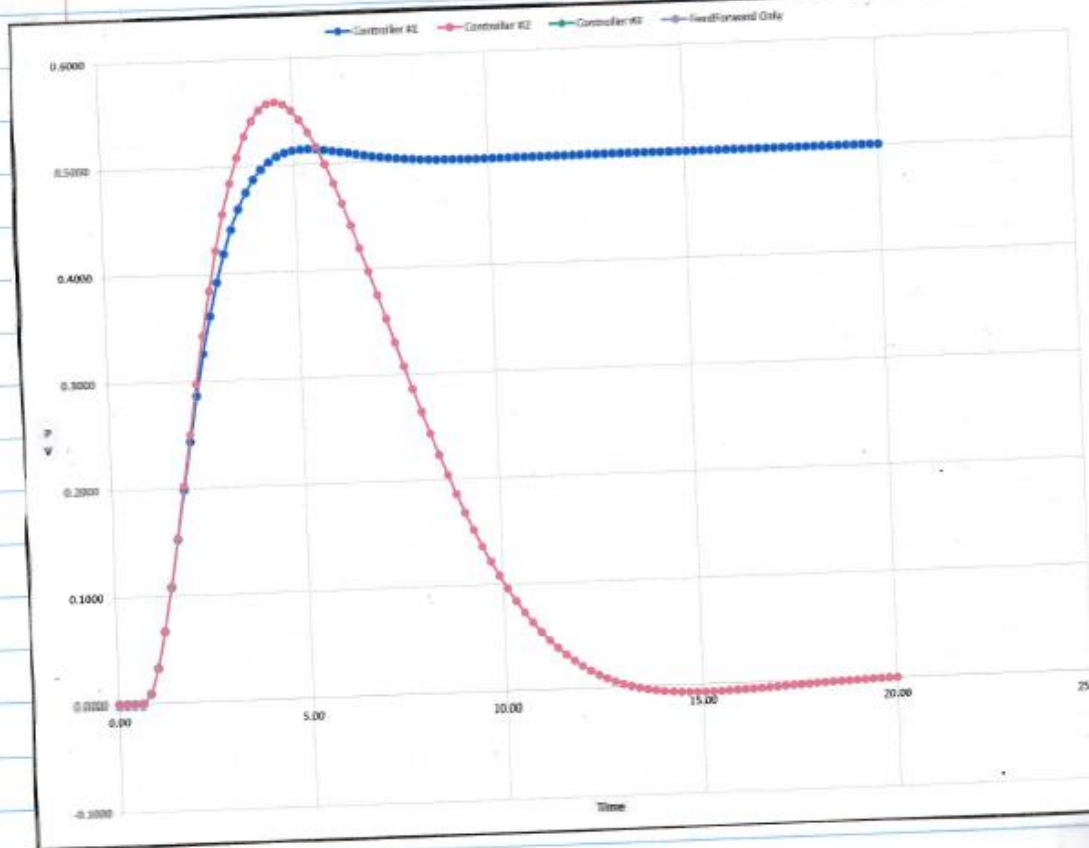
$$\tau_I = 2.125$$

Running the simulation with exactly the same input as in part D but now with the new controller we obtain the graph shown below

We observe that the PI rejects the disturbance nicely with no offset and a small overshoot

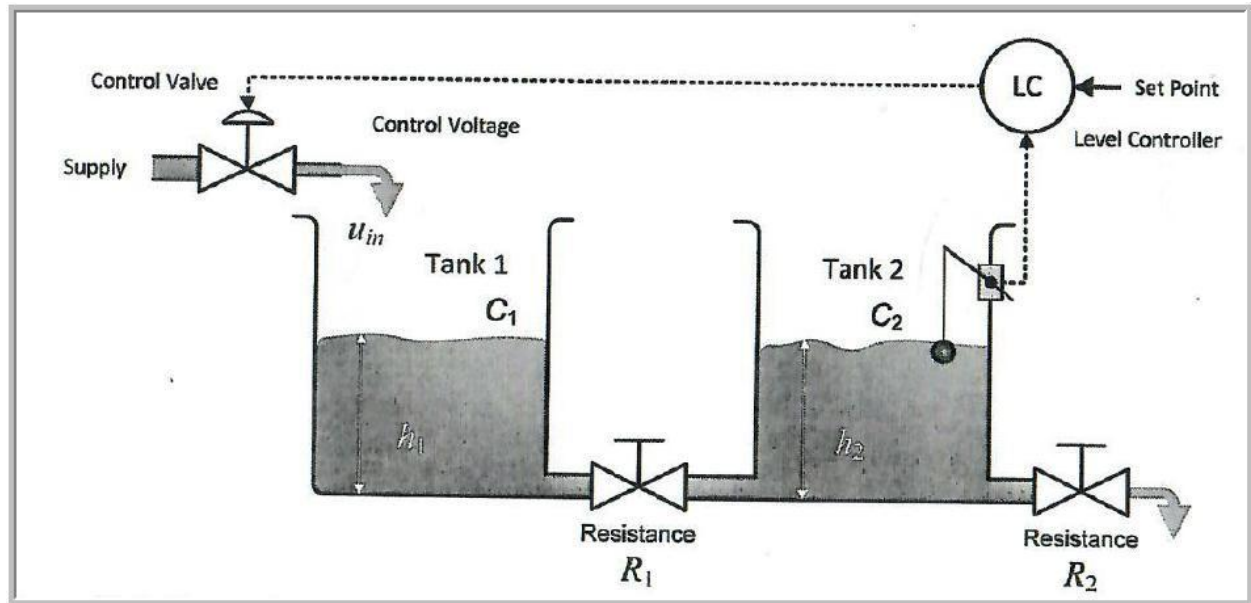
The settling time is around 12 min. For faster settling time we may consider adding some derivative action. If the overshoot is a problem we may consider implementing a Feed forward controller since the effect of the disturbance on the tank temperature is well understood.

Plots of PD Controller (Blue Line) with offset and PI Controller (Red line) with the offset eliminated.



Problem: Two-Tank Level Control

Very often in the chemical process industries two-tank level control is used. The first tank receives a liquid stream and passes the liquid to the second tank as shown in the diagram below.



A level transmitter is located in the second tank and provides the required feedback for a PID controller which regulates the level in the second tank. This configuration allows large changes in the first tank while minimizing disturbances in the second tank.

The capacitance of the first and second tank is $C_1=C_2=1.0 \text{ m}^3/\text{m}$ and the resistances out of the first and second tank $R_1=0.5 \text{ m}/(\text{m}^3/\text{min})$ and $R_2=0.333 \text{ m}/(\text{m}^3/\text{min})$. The volumetric flow rate through the gate valves varies linearly with the pressure drop. Assume that the time delays in the control loop including the transportation lag between the first and second tank amount to around 35% of the dominant time constant of the system.

1. Derive the process and disturbance transfer functions assuming that the inlet flow acts also as a disturbance
2. Design a PID controller with $\text{GM}=3$ and $\text{PM}=60$ degrees and perform a simulation for set point and disturbance change of unit size.
3. Compare your controller with:
 - a. Ziegler-Nichols Open Loop
 - b. Cohen-Coon
 - c. Ziegler-Nichols Closed Loop
 - d. Luyben-Tyreus
 - e. IMC
 - f. Your own personal design

Present the comparison results in a table format. In the table calculate the following quantities both for set point tracking and disturbance rejection of unit size.

1. Overshoot (if present)
2. Settling time
3. Gain Margin
4. Phase Margin
5. Maximum amount of time delay allowed before the onset of instability
6. Peak Value of the sensitivity function
7. Peak Value of the complimentary sensitivity function

Include all the simulation graphs in your final report along with the Excel table shown below

	GM=3 PM=60 deg	Ziegler-Nichols Open Loop	Cohen-Coon	Ziegler-Nichols Closed Loop	Luyben-Tyreus	IMC	My Personal Controller	Selected Controller
Overshoot (if present)								
Settling time								
Gain Margin								
Phase Margin								
Maximum amount of time delay allowed before instability								
Peak Value of the sensitivity function								
Peak Value of the complimentary sensitivity function								
Controller Gain, K_c								
Controller Reset Time, τ_i								
Controller Derivative Time, τ_D								

Solution: Two-Tank Level Control

1. Dynamic Model and Transfer-Function Development

The process considered in this problem consists of two interacting tanks in series, with the controlled variable being the liquid level in the second tank. The inlet stream enters Tank 1 through a control valve, flows from Tank 1 to Tank 2 through a hydraulic resistance, and finally exits Tank 2 through a second resistance. A level transmitter in Tank 2 measures the liquid height and provides feedback to the level controller. Because the second tank is the controlled vessel, the objective of the modeling step is to obtain both the process transfer function, relating the manipulated inlet flow to the level in Tank 2, and the disturbance transfer function, relating disturbance changes in inlet flow to the same output. Since the process statement specifies that the inlet flow should also be treated as a disturbance, the inlet stream is modeled as the sum of a manipulated component and a disturbance component.

Let $h_1(t)$ and $h_2(t)$ denote the liquid levels in Tanks 1 and 2, respectively, in meters. The two tank capacitances are given as $C_1 = C_2 = 1.0 \text{ m}^3/\text{m}$, the hydraulic resistance between Tanks 1 and 2 is $R_1 = 0.5 \text{ m}/(\text{m}^3/\text{min})$, and the outlet resistance from Tank 2 is $R_2 = 0.333 \text{ m}/(\text{m}^3/\text{min})$. The inlet flow is written as the sum of a manipulated input and a disturbance input,

$$q_{in}(t) = u(t) + d(t)$$

where $u(t)$ is the manipulated flow contribution and $d(t)$ is the disturbance contribution, both in m^3/min . Because the gate-valve flows vary linearly with pressure drop, the flow from Tank 1 to Tank 2 is proportional to the level difference across R_1 , and the outlet flow from Tank 2 is proportional to the level in Tank 2. Thus,

$$q_{12}(t) = \frac{h_1(t) - h_2(t)}{R_1}$$

$$q_2(t) = \frac{h_2(t)}{R_2}$$

Using these relations, the dynamic mass balance on Tank 1 is

$$C_1 \frac{dh_1}{dt} = q_{in} - q_{12}$$

or, after substitution,

$$C_1 \frac{dh_1}{dt} = u + d - \frac{h_1 - h_2}{R_1}$$

Similarly, the dynamic mass balance on Tank 2 is

$$C_2 \frac{dh_2}{dt} = q_{12} - q_2$$

which becomes

$$C_2 \frac{dh_2}{dt} = \frac{h_1 - h_2}{R_1} - \frac{h_2}{R_2}$$

These two first-order equations form the complete dynamic model of the interacting two-tank system in the time domain.

To derive the transfer functions, the model is written in deviation-variable form and transformed to the Laplace domain assuming zero initial conditions. The Tank 1 balance becomes

$$C_1 s H_1(s) = U(s) + D(s) - \frac{H_1(s) - H_2(s)}{R_1}$$

and the Tank 2 balance becomes

$$C_2 s H_2(s) = \frac{H_1(s) - H_2(s)}{R_1} - \frac{H_2(s)}{R_2}$$

Rearranging both equations gives

$$\begin{aligned} \left(C_1 s + \frac{1}{R_1}\right) H_1(s) - \frac{1}{R_1} H_2(s) &= U(s) + D(s) \\ -\frac{1}{R_1} H_1(s) + \left(C_2 s + \frac{1}{R_1} + \frac{1}{R_2}\right) H_2(s) &= 0 \end{aligned}$$

The second equation may be used to eliminate $H_1(s)$. Solving for $H_2(s)$ in terms of the combined input $U(s) + D(s)$ gives

$$H_2(s) = \frac{\frac{1}{R_1}}{\left(C_1 s + \frac{1}{R_1}\right) \left(C_2 s + \frac{1}{R_1} + \frac{1}{R_2}\right) - \frac{1}{R_1^2}} [U(s) + D(s)]$$

From this result, the process and disturbance transfer functions follow directly. The process transfer function is defined as the response of $H_2(s)$ to the manipulated input $U(s)$ with the disturbance held at zero:

$$G_p(s) = \frac{H_2(s)}{U(s)} = \frac{\frac{1}{R_1}}{\left(C_1 s + \frac{1}{R_1}\right) \left(C_2 s + \frac{1}{R_1} + \frac{1}{R_2}\right) - \frac{1}{R_1^2}}$$

The disturbance transfer function is defined as the response of $H_2(s)$ to the disturbance $D(s)$ with the manipulated input held at zero:

$$G_d(s) = \frac{H_2(s)}{D(s)} = \frac{\frac{1}{R_1}}{\left(C_1 s + \frac{1}{R_1}\right) \left(C_2 s + \frac{1}{R_1} + \frac{1}{R_2}\right) - \frac{1}{R_1^2}}$$

For this particular problem, the manipulated inlet flow and the disturbance inlet flow enter the first tank through the same location and in the same units, so the process and disturbance transfer functions have the same form. Expanding the denominator gives a more compact expression:

$$G_p(s) = G_d(s) = \frac{\frac{1}{R_1}}{C_1 C_2 s^2 + \left[C_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + C_2 \left(\frac{1}{R_1} \right) \right] s + \frac{1}{R_1 R_2}}$$

Substituting the given numerical values $C_1 = C_2 = 1.0$, $R_1 = 0.5$, and $R_2 = 0.333$ yields

$$\frac{1}{R_1} = 2 \text{ and } \frac{1}{R_2} \approx 3.00$$

so the transfer functions become approximately

$$G_p(s) = G_d(s) \approx \frac{2}{s^2 + 7s + 6}$$

or, in factored form,

$$G_p(s) = G_d(s) \approx \frac{2}{(s + 1)(s + 6)}$$

This form is especially useful because it makes the process time scales clear. The two poles are at $s = -1$ and $s = -6$, so the dominant pole is the slower one at $s = -1$. Therefore, the dominant time constant of the two-tank system is approximately

$$\tau_{dom} = 1 \text{ min}$$

The process statement also specifies that the total loop time delay, including the transportation lag between the first and second tanks, is approximately 35% of the dominant time constant. Using the dominant time constant just obtained, the effective delay is

$$\theta = 0.35\tau_{dom} = 0.35(1) = 0.35 \text{ min}$$

Including this delay, the final transfer functions for the process and disturbance become

$$G_p(s) \approx \frac{2e^{-0.35s}}{(s + 1)(s + 6)}$$

$$G_d(s) \approx \frac{2e^{-0.35s}}{(s + 1)(s + 6)}$$

Which can be simplified to

$$G_p(s) \approx \frac{0.333 e^{-0.35s}}{(1.0s + 1)(0.1667s + 1)}$$

$$G_d(s) \approx \frac{0.333 e^{-0.35s}}{(1.0s + 1)(0.1667s + 1)}$$

These expressions complete the dynamic-model development required in Part 1. The process transfer function describes how changes in the manipulated inlet flow affect the level in Tank 2, while the disturbance transfer function describes how undesired inlet-flow variations propagate to the same controlled variable. Both transfer functions show that the system behaves as a stable, second-order interacting process with a small but important time delay. This model provides the basis for the controller design and comparison carried out in the later parts of the report.

2. Controller Design for $GM = 3$ and $PM = 60^\circ$, and Closed-Loop Simulation

The first controller designed for this process was a feedback controller required to satisfy a gain margin of 3 and a phase margin of 60° . This controller was developed first because it directly answered the requirement of Part 2 and also established a balanced reference case for the broader comparison performed later. A PI controller was selected for this part for two reasons. First, Chapter 6 identifies PI control as the most commonly used mode for pressure, flow, and level control, while derivative action is more commonly associated with slower, low-noise loops such as temperature control. Second, the numerical design problem in Part 2 involved two robustness targets, $GM = 3$ and $PM = 60^\circ$, so it was natural to solve for two unknown controller parameters only, namely K_c and τ_I . By setting the derivative time equal to zero, the design problem became well posed for the Newton method.

The process model used in this part was the exact second-order-plus-time-delay transfer function derived previously for the two-tank system,

$$G_p(s) = \frac{0.333 e^{-0.35s}}{(s + 1)(0.1667s + 1)}$$

This form makes the process constants easier to interpret. The overall process gain is $K_p = 0.333$, the dominant lag is $\tau_1 = 1.0$ min, the second lag is $\tau_2 = 0.1667$ min, and the transportation delay is $\theta = 0.35$ min. The disturbance path has the same dynamic structure, so the same process constants were used there as well. Since this controller was designed directly from gain-margin and phase-margin targets, no model simplification was required in this part.

With derivative action omitted, the controller transfer function used here was

$$G_c(s) = K_c \left(1 + \frac{1}{\tau_I s} \right)$$

so that

$$\tau_D = 0$$

The unknown controller parameters were therefore the controller gain K_c and the reset time τ_I . These were determined using the **BODE Stability Analysis** sheet in the CHE Calculator. For this analysis, the process was entered in the second-order form used by the CHE Calculator, with overall gain $K_p = 0.333$, time delay $\theta = 0.35$ min, and equivalent second-order parameters defined from the two first-order lags as

$$\tau = \sqrt{\tau_1 \tau_2}$$

and

$$\zeta = \frac{\tau_1 + \tau_2}{2\tau}$$

Using $\tau_1 = 1.0$ min and $\tau_2 = 0.1667$ min gave the values of τ and ζ entered into the Bode sheet. A PI controller was used in this part both because PI control is a natural choice for a level-control process and because the design problem contained only two unknown controller parameters. The

controller was represented in non-interactive PI form, and the sheet then generated the amplitude-ratio and phase-angle data for the open-loop transfer function over a frequency range.

Two frequencies are central to this analysis. The **critical frequency** ω_c is the frequency at which the phase angle equals -180° , and the **gain crossover frequency** ω_g is the frequency at which the amplitude ratio equals one. The gain margin depends on the amplitude ratio evaluated at the critical frequency, while the phase margin depends on the phase angle evaluated at the gain crossover frequency. By default, however, the Bode sheet does not update ω_c and ω_g automatically. Those cells are blank and must otherwise be read manually from the tabulated columns and entered by hand. That creates a serious problem if one attempts to use Newton's method to solve for K_c and τ_I , because the controller constants would be changing while the frequencies needed to evaluate GM and PM would remain fixed. In that case, the Newton solver would either fail to converge or converge to incorrect values.

For this reason, interpolation functions were written into the Bode table so that ω_c and ω_g were updated automatically as the controller constants changed. The critical frequency was obtained by interpolating between the two neighboring rows that bracketed the condition $PA = -180^\circ$, while the gain crossover frequency was obtained by interpolating between the two neighboring rows that bracketed the condition $AR = 1$. This step was essential because it tied the frequency calculations directly to the changing controller parameters. Once these interpolation functions were in place, the critical-frequency and gain-crossover equations were effectively satisfied automatically, since the interpolation itself located the frequencies that solved those crossing conditions. As a result, the Newton method only needed to solve for the two controller parameters K_c and τ_I , which matched the two robustness targets in the problem.

The Newton method was then used to vary K_c and τ_I until the desired conditions

$$GM = 3 \text{ and } PM = 60^\circ$$

were met. The resulting controller constants were

$$K_c = 3.65$$

$$\tau_I = 1.1322 \text{ min}$$

$$\tau_D = 0$$

These values gave the required gain margin and phase margin to the desired precision. The same Bode sheet also provided the maximum additional delay before the onset of instability, along with the peak sensitivity and peak complementary sensitivity, which were recorded for use later in the controller comparison.

After the controller constants had been determined, the time-domain response of the closed loop was evaluated using the **Automation Panel**. In this panel, the same exact process model was retained. The overall gain remained 0.333, the dominant lag remained 1.0 min, the process time delay remained 0.35 min, and the additional lag of 0.1667 min was entered as a sensor lag. This was done because the Automation Panel provides a convenient separate location for an additional

first-order lag through the sensor block, allowing the second dynamic lag in the process model to be represented explicitly without changing the dominant process lag or the actual time delay. The same process structure was used for the disturbance path, so both the manipulated input and disturbance input were evaluated on the same plant.

Two responses were generated. First, a **unit set-point change** was imposed while the disturbance was held at zero. Second, a **unit disturbance change** was imposed while the set point was held constant. The controller constants K_c , τ_I , and τ_D obtained from the Bode design were entered directly into the controller block, and the Automation Panel then computed the corresponding process-variable response.

The resulting responses are shown in Figure 1 and Figure 2. **Figure 1** presents the unit set-point response of the $GM = 3$, $PM = 60^\circ$ PI controller. The process variable rises to its target with only a modest overshoot and then returns smoothly to steady state, indicating a balanced compromise between speed and damping. **Figure 2** presents the corresponding unit disturbance response. In this case, the controller rejects the imposed load change and returns the process variable to zero deviation without steady-state offset. Taken together, these two responses show that the controller designed directly from robustness targets provides stable and well-damped closed-loop behavior and therefore serves as an appropriate reference case for the tuning-method comparison in Part 3.

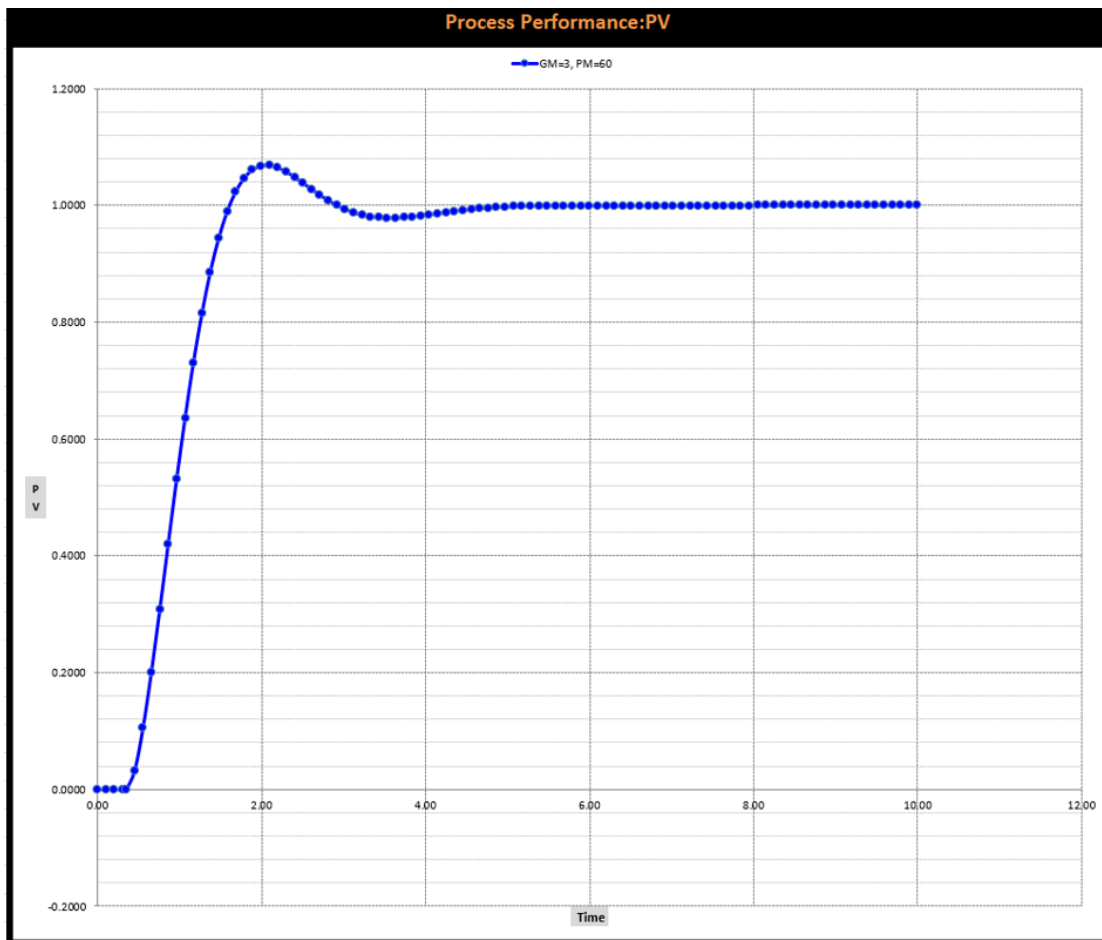


Figure 1: Closed-loop responses of the $GM = 3$, $PM = 60^\circ$ controller for a set-point change

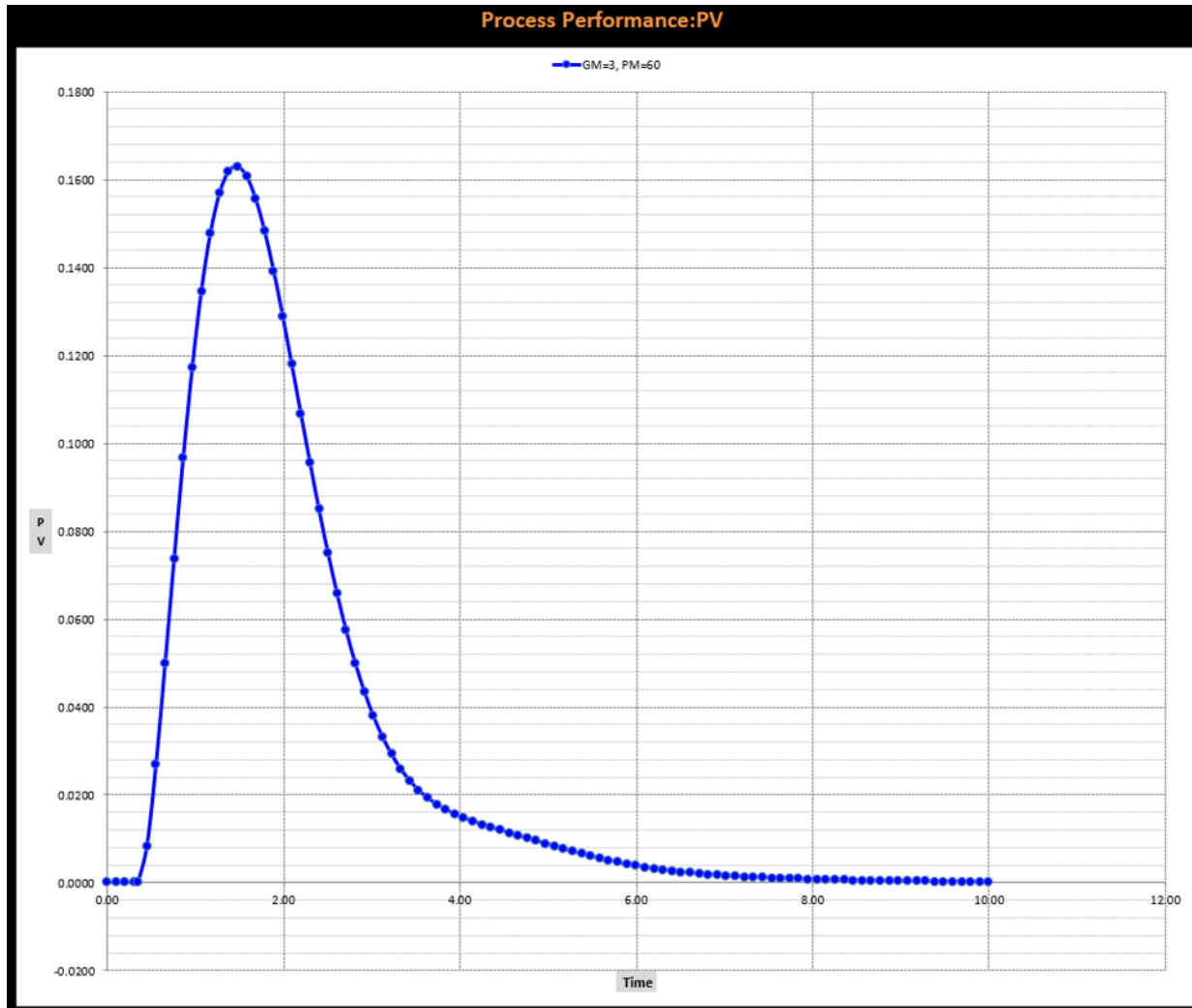


Figure 2: Closed-loop responses of the $GM = 3$, $PM = 60^\circ$ controller for disturbance change

In summary, this part began with the exact process model, selected PI control both because it is a natural choice for level control and because it provided the correct number of unknowns for the robustness design problem, introduced interpolation functions so that the critical frequency and gain crossover frequency would update automatically during the Newton solution, solved for the controller constants K_c and τ_I from the required gain and phase margins, and then evaluated the resulting controller in the Automation Panel for both set-point and disturbance changes. The outcome was a controller that met the specified robustness requirements while also giving a stable and well-damped closed-loop response.

3. Comparison of Standard Tuning Methods and Selection of a Final Controller

After completing the $GM = 3$, $PM = 60^\circ$ design in Part 2, the next objective was to compare that controller against several standard tuning methods from Chapter 6. The controllers examined were Ziegler–Nichols Open Loop, Cohen–Coon, Ziegler–Nichols Closed Loop, Tyreus–Luyben, IMC, and a personal controller design. The $GM = 3$, $PM = 60^\circ$ controller from Part 2 was retained as the reference case. The comparison was carried out in two parts. First, the controller constants were determined using the appropriate tuning relation for each method. Second, each controller was evaluated on the same plant using the Automation Panel and the BODE Stability Analysis sheet. This ensured that the comparison reflected differences in controller tuning rather than differences in the process model used for testing.

The exact plant used for the final comparison remained

$$G_p(s) = \frac{2e^{-0.35s}}{(s+1)(s+6)}$$

or, equivalently,

$$G_p(s) = \frac{0.333 e^{-0.35s}}{(s+1)(0.1667s+1)}$$

with the same model used for the disturbance path. In the Automation Panel and Bode sheet, the dominant lag was entered as the process lag, the transportation delay was kept explicitly as the process time delay, and the smaller second lag was entered as an additional sensor lag for the Automation Panel and as a second process delay (labelled τ_3) in the Bode sheet. This preserved the exact second-order-plus-time-delay structure of the original process during the final controller comparison.

3.1 Controller-derivation strategy

Not all controller constants were obtained in the same way. This distinction is important because some Chapter 6 tuning formulas are written for first-order-plus-time-delay process models, while others are based on the ultimate gain and ultimate period or on direct robustness targets.

The $GM = 3$, $PM = 60^\circ$ controller from Part 2 was obtained directly from the exact process model using the Bode sheet. Since two robustness targets were specified and only two controller parameters were left unknown, a PI controller was used for that design. This controller therefore required no process approximation.

Three controllers in Part 3 did require a first-order-plus-time-delay approximation for the purpose of calculating their tuning constants:

1. Ziegler–Nichols Open Loop
2. Cohen–Coon
3. IMC / Direct Synthesis

This follows directly from Chapter 6. The Ziegler–Nichols Open Loop method is explicitly developed for an empirical FOPTD model, and the chapter states that the open-loop Ziegler–

Nichols rules were developed for self-regulating processes described by that form. Cohen–Coon is also presented in the chapter as a tuning method developed for an FOPTD process and for a non-interactive controller. Direct Synthesis and IMC are introduced in Chapter 6 as more general methods, but the specific tuning relations used here are the first-order-and-time-delay forms presented in the chapter.

The remaining controllers did not require an FOPTD approximation during the tuning step:

1. Ziegler–Nichols Closed Loop
2. Tyreus–Luyben
3. Personal Controller

The closed-loop Ziegler–Nichols method and the Tyreus–Luyben method are based on the ultimate gain K_u and ultimate period P_u , not on an FOPTD fit. Chapter 6 presents those methods in that form. The personal controller was chosen directly and therefore also required no process approximation.

3.2 FOPTD approximation using Skogestad’s half-rule

For the FOPTD-based methods, the original second-order-plus-time-delay process was approximated by a first-order-plus-time-delay model using Skogestad’s half-rule. The exact process has a dominant lag $\tau_1 = 1.0$ min, a smaller lag $\tau_2 = 0.1667$ min, and a time delay $\theta = 0.35$ min. Under Skogestad’s approximation, the reduced constants become

$$\begin{aligned} K_p &= 0.333 \\ \tau &= \tau_1 + \frac{\tau_2}{2} = 1.0833 \\ \theta^* &= \theta + \frac{\tau_2}{2} = 0.4333 \end{aligned}$$

so the corresponding first-order-plus-time-delay approximation is

$$G_p(s) \approx \frac{0.333 e^{-0.4333s}}{1.0833s + 1}$$

These reduced constants were used only to calculate the controller parameters for the Ziegler–Nichols Open Loop, Cohen–Coon, and IMC methods. They were **not** used as the final plant model in the controller comparison. After the controller constants were obtained, all controllers were evaluated on the same full process model so that the comparison reflected differences in tuning rather than differences in the plant representation.

3.3 Calculation of controller constants

The controller constants were not all obtained in the same way. Three of the tuning methods—Ziegler–Nichols Open Loop, Cohen–Coon, and IMC—required the first-order-plus-time-delay approximation introduced in the previous section, since their tuning relations are written in terms of the three reduced process constants K_p , τ , and θ . The remaining controllers were obtained either directly from the exact model, from the ultimate gain and period, or by direct selection. Once the constants were calculated, however, every controller was evaluated on the same full plant model so that the comparison would reflect only differences in tuning.

The $GM = 3$, $PM = 60^\circ$ controller from Part 2 was retained without modification. It was implemented as a PI controller, both because PI control is a natural choice for a level-control process and because the design problem in Part 2 involved only two unknown controller parameters. The controller constants for that case had already been obtained directly from the exact model by solving for the values of K_c and τ_I that satisfied the two target robustness conditions.

For the Ziegler–Nichols Open Loop controller, the tuning constants were calculated from the empirical open-loop PID relations. These are

$$\begin{aligned} K_c &= \frac{1.2}{\alpha K_p} \frac{\tau}{\theta} \\ \tau_I &= 2\theta \\ \tau_D &= \frac{\theta}{2} \end{aligned}$$

where α is a safety-margin factor. In this problem, a value of $\alpha = 1$ was used, corresponding to the original Ziegler–Nichols correlation without any additional conservatism.

For the Cohen–Coon controller, the PID tuning relations used were

$$\begin{aligned} K_c &= \frac{1.35}{\alpha K_p} \left(\frac{\tau}{\theta} + 0.185 \right) \\ \tau_I &= 2.5\theta \left(\frac{\tau + 0.185\theta}{\tau + 0.611\theta} \right) \\ \tau_D &= 0.37 \left(\frac{\tau\theta}{\tau + 0.185\theta} \right) \end{aligned}$$

Again, $\alpha = 1$ was used so that the original Cohen–Coon correlation was applied directly.

For the IMC controller, the constants were obtained from the first-order-and-time-delay IMC tuning relation. The equations used were

$$\begin{aligned} K_c &= \frac{\tau + \theta/2}{K_p(\tau_c + \theta/2)} \\ \tau_I &= \tau + \frac{\theta}{2} \\ \tau_D &= \frac{\tau\theta/2}{\tau + \theta/2} \end{aligned}$$

where τ_c is the selected design parameter for the IMC tuning. This method gave a controller that was expected to be less aggressive than the classical empirical correlations and more balanced in terms of robustness.

For the Ziegler–Nichols Closed Loop controller, the tuning constants were obtained from the ultimate gain K_u and ultimate period P_u , which were approximated using the controller from Part 2. Using those values, the controller constants were calculated from

$$K_c = 0.60K_u$$

$$\begin{aligned}\tau_I &= 0.50P_u \\ \tau_D &= 0.125P_u\end{aligned}$$

The Tyreus–Luyben controller was obtained from the same K_u and P_u values, but with the more conservative relations

$$\begin{aligned}K_c &= 0.45K_u \\ \tau_I &= 2.2P_u \\ \tau_D &= 0.1587P_u\end{aligned}$$

Because these formulas are more conservative than the Ziegler–Nichols Closed Loop constants, the resulting controller was expected to have a slower but more robust response.

Finally, the personal controller was selected directly rather than from a published tuning correlation. It was intentionally chosen as a PI controller with

$$K_c = 4.0 \text{ and } \tau_I = 1.2 \text{ min}$$

This choice was made to provide a practical comparison case against the formal tuning methods. The values were selected to give a smooth and well-damped level response without introducing derivative action, which would likely show limited benefit for this process.

In summary, the controller constants were obtained in three different ways. The $GM = 3$, $PM = 60^\circ$ controller was carried over directly from the robustness design in Part 2. The Ziegler–Nichols Open Loop, Cohen–Coon, and IMC controllers were calculated from the FOPTD approximation using the corresponding tuning relations. The Ziegler–Nichols Closed Loop and Tyreus–Luyben controllers were calculated from the ultimate gain and period, and the personal controller was selected directly as a practical PI design. Once these constants were determined, every controller was evaluated on the same full process model.

3.4 Time-domain evaluation using the Automation Panel

After the controller constants had been determined, each controller was evaluated in the time domain using the Automation Panel. This step was the same for every controller. Only the controller constants K_c , τ_I , and τ_D were changed from one case to another; the plant model itself remained fixed. This is important, because it ensured that any differences in the response plots were caused by the controller tuning alone and not by changes in the process representation.

For all controllers, the same full plant model was used in the Automation Panel. The overall process gain remained 0.333, the dominant lag remained 1.0 min, the transportation delay remained 0.35 min, and the smaller second lag of 0.1667 min was entered separately as a sensor lag. This allowed the complete plant dynamics to be retained during the comparison, even for the controllers whose constants had originally been obtained from an FOPTD approximation. The same process structure was used for both the manipulated-input path and the disturbance path.

For each controller, two time-domain responses were generated:

1. a unit set-point change, with the disturbance held at zero
2. a unit disturbance change, with the set point held constant

The resulting plots were then used to compare the transient behavior of the controllers. The numerical performance measures recorded from these plots were overshoot and settling time. These values were taken directly from the Automation Panel output and then summarized in the comparison table.

One important detail in interpreting the disturbance responses is that the numerical table reports only the overshoot, not the undershoot. This means that a controller with little positive overshoot may still appear better numerically than it actually is if its response contains a large negative excursion before returning to zero. This effect was especially noticeable for the more conservative controllers. For that reason, disturbance settling time was treated as the more meaningful disturbance-performance indicator, while the appendix plots were used to show the full shape of the transient response.

3.5 Robustness analysis using the Bode stability sheet

In addition to the time-domain comparison, each controller was also evaluated in the frequency domain using the BODE Stability Analysis sheet. The purpose of this analysis was to compare the controllers not only in terms of performance, such as overshoot and settling time, but also in terms of robustness, that is, how much uncertainty or change in the loop can be tolerated before the controller approaches instability. The main robustness quantities considered were the gain margin, phase margin, peak sensitivity function, and peak complementary sensitivity function.

For each controller, the corresponding values of K_c , τ_I , and τ_D were entered into the Bode sheet while the plant model remained fixed. The sheet then generated the open-loop amplitude ratio and phase angle over a range of frequencies. The two key frequencies in this analysis were the critical frequency ω_c and the gain crossover frequency ω_g . The critical frequency is defined by the condition

$$PA(\omega_c) + 180 = 0$$

while the gain crossover frequency is defined by

$$AR(\omega_g) - 1 = 0$$

As in Part 2, interpolation functions were used to identify ω_c and ω_g directly from the Bode table. Because the interpolation formulas locate the frequencies that satisfy these two crossing conditions, the corresponding equations evaluate to zero automatically once the formulas are written correctly. This made additional Newton iteration on ω_c and ω_g unnecessary.

The gain margin was then calculated from the amplitude ratio at the critical frequency,

$$GM = \frac{1}{AR(\omega_c)}$$

and the phase margin was evaluated at the gain crossover frequency. The maximum additional time delay that the loop could tolerate before becoming unstable was calculated from

$$\Delta\theta_{\max} = \frac{PM}{\omega_g} \frac{\pi}{180}$$

Figure 3: Robustness Analysis through Gain Ratio and Dead Time Ratio

Taken together, the gain margin, phase margin, allowable additional delay, peak sensitivity, peak complementary sensitivity, and gain-ratio/dead-time-ratio plot gave a much more complete picture of controller quality than the time-domain response alone. These measures made it possible to distinguish between controllers that were merely fast and controllers that were genuinely robust.

3.6 Overall comparison of all controllers

The full comparison of time-domain performance and robustness is summarized in Table 1, while the gain-ratio versus dead-time-ratio robustness plot is shown separately in Figure 6

Table 1: Comparison of controller performance and robustness

		GM=3 PM=60 deg	Ziegler- Nichols Open Loop	Cohen-Coon	Ziegler-Nichols Closed Loop	Luyben-Tyresus	IMC	My Personal Controller	Selected Controller IMC
Overshoot (if present)	Set-point	0.068	0.478	0.703	0.183	0.000	0.100	0.084	0.100
	Disturbance	0.163	0.108	0.109	0.115	0.122	0.125	0.159	0.125
Settling time, min	Set-point	3.84	3.02	5.48	3.64	17.54	3.01	4.03	3.01
	Disturbance	3.64	1.79	2.92	2.71	13.04	3.42	3.63	3.42
Gain Margin		3.00	1.70	1.44	2.21	2.45	2.61	2.78	2.61
Phase Margin, deg		60.00	42.44	26.36	58.88	101.98	62.72	58.93	62.72
Maximum amount of time delay allowed before the onset of instability, min		0.922	0.286	0.149	0.556	1.361	0.653	0.840	0.653
Peak Value of the sensitivity function		1.670	2.563	3.758	1.882	1.721	1.716	1.742	1.716
Peak Value of the complimentary sensitivity function		1.002	1.678	2.959	1.067	0.999	1.000	1.036	1.000
Controller Gain, Kc		3.65	9.01	10.88	6.57	4.92	6.01	4.00	6.01
Controller Reset Time, τ_i		1.132	0.867	0.997	0.985	4.335	1.30	1.20	1.30
Controller Derivative Time, τ_D		0.000	0.217	0.149	0.246	0.313	0.181	0.000	0.181

The $GM = 3$, $PM = 60^\circ$ controller from Part 2 served as the reference case. As expected, it gave a balanced response with small overshoot, reasonable settling time, and robustness values that were good by construction. It did not dominate every individual category, but it performed consistently well in both the time-domain and frequency-domain measures. This made it a useful benchmark for the remaining controllers.

The Ziegler–Nichols Open Loop controller was clearly aggressive. Its set-point response exhibited substantial overshoot and oscillation, and its robustness values were comparatively weak. In particular, its gain margin and phase margin were both lower than those of the more balanced controllers, and its sensitivity measures were noticeably higher. This is consistent with the well-known tendency of the original Ziegler–Nichols Open Loop method to produce fast but oscillatory responses.

The Cohen–Coon controller was also aggressive and, in this study, performed even more poorly overall than Ziegler–Nichols Open Loop. Its set-point response showed the largest overshoot and the strongest oscillatory behavior in the comparison. Its robustness values were also among the weakest, with low stability margins and the highest sensitivity measures. Although the response was fast, it was clearly too aggressive for this process and offered little robustness against model uncertainty or additional delay.

The Ziegler–Nichols Closed Loop controller represented an improvement over the two open-loop empirical methods. Its response was still somewhat aggressive, but it was better damped and less oscillatory than either Ziegler–Nichols Open Loop or Cohen–Coon. Its robustness values were also noticeably stronger, placing it in a more acceptable range both in Table 1 and in the robustness plot.

The Tyreus–Luyben controller was by far the most conservative. This is apparent in both the time-domain plots and the robustness analysis. Its gain margin, phase margin, and delay tolerance were all very large, and on the gain-ratio versus dead-time-ratio plot in Figure 6, it extended so far into the robust region that it did not fit neatly within the graph window. However, that conservatism came at the expense of speed. Its set-point and disturbance responses were significantly slower than the others, making it less attractive in practice despite its excellent robustness.

The IMC controller provided the best overall compromise. Its time responses were smooth and well damped, with very small overshoot and reasonable settling times for both the set-point and disturbance cases. At the same time, its robustness values remained strong. Its gain and delay ratios placed it comfortably within the robust region of Figure 6, and its peak sensitivity and peak complementary sensitivity were among the lowest in the comparison. Even though its tuning constants were obtained from an FOPTD approximation, it performed very well when evaluated on the full plant, which is what ultimately matters in the final comparison.

The personal controller also performed well. It gave a smooth response and strong robustness and was competitive with the IMC controller in several categories. As a PI controller, it also supported the observation that derivative action was not especially important for this delayed level-control process. However, the IMC controller still gave the better overall balance of damping, settling behavior, and robustness, particularly when both the time-domain responses and the frequency-domain measures were considered together.

Taken together, the results show a clear pattern. The most aggressive empirical controllers, namely Cohen–Coon and Ziegler–Nichols Open Loop, produced fast but oscillatory behavior and weak robustness. The most conservative controller, Tyreus–Luyben, gave excellent robustness but was much too slow. The most balanced controllers were IMC, the personal controller, and the $GM = 3$, $PM = 60^\circ$ controller, with IMC giving the strongest overall combination of performance and robustness.

3.7 Final controller selection and ranking

Based on the complete comparison of time-domain behavior and robustness, the IMC controller was selected as the final recommended controller. It provided the best overall compromise between closed-loop performance and robustness. Its set-point response was smooth and well damped, its disturbance rejection remained acceptable, and its robustness measures were consistently strong. In particular, its gain margin, phase margin, allowable additional delay, gain-ratio/dead-time-ratio position, and sensitivity measures all indicated a controller that was both stable and reasonably tolerant to plant uncertainty.

The personal controller was the strongest alternative to IMC. It performed well in both the time-domain and frequency-domain comparisons and gave a smooth response without excessive aggressiveness. However, IMC still provided the more balanced overall result when both performance and robustness were considered together. The $GM = 3$, $PM = 60^\circ$ controller also performed well and served as a strong reference case, but it did not match the overall balance achieved by IMC.

The Tyreus–Luyben controller was the most robust in the strictest sense, particularly with respect to gain and delay variation, but it was too conservative in practice. Its responses were significantly slower than the others, which reduced its usefulness despite its excellent margins.

The Ziegler–Nichols Closed Loop controller represented a reasonable improvement over the open-loop empirical methods, but it was still more aggressive and less balanced than the top-performing alternatives.

The two weakest controllers in the comparison were Ziegler–Nichols Open Loop and Cohen–Coon. Both produced fast responses, but they also showed much larger oscillations and weaker robustness than the other candidates. Of the two, Cohen–Coon was the least desirable overall because it combined strong oscillation with the poorest robustness metrics.

The controllers were therefore ranked from best to worst as follows:

1. IMC
2. Personal Controller
3. $GM = 3$, $PM = 60^\circ$ PI Controller
4. Tyreus–Luyben
5. Ziegler–Nichols Closed Loop
6. Ziegler–Nichols Open Loop
7. Cohen–Coon

An additional observation is worth noting. Although several of the classical methods were implemented in PID form whenever possible, the two PI controllers—the $GM = 3$, $PM = 60^\circ$ controller and the personal controller—also performed very well. This suggests that derivative action was not especially beneficial for this delayed level-control process and that a well-tuned PI controller was already capable of providing strong overall performance.

Conclusion

The comparison in Part 3 showed that the choice of tuning method has a major effect on both closed-loop performance and robustness for the two-tank level-control process. Some methods required a first-order-plus-time-delay approximation to obtain their controller constants, while others were based directly on the full process model or on the ultimate gain and ultimate period. Regardless of how the constants were obtained, all controllers were ultimately evaluated on the same plant, which made the comparison fair and consistent.

Among the controllers considered, the IMC controller gave the best overall balance of smooth set-point tracking, acceptable disturbance rejection, and strong robustness. The more aggressive

empirical methods, particularly Cohen–Coon and Ziegler–Nichols Open Loop, were fast but excessively oscillatory and less robust. The most conservative method, Tyreus–Luyben, provided excellent robustness but at the cost of very slow response. The personal controller and the $GM = 3$, $PM = 60^\circ$ PI controller both performed well, which also suggested that derivative action was not especially important for this process.

Overall, the IMC controller was selected as the final recommended controller because it achieved the strongest compromise between performance and robustness for the two-tank level-control system.

Appendix I: Additional Graphs and Code

```
Function OmegaAtAR(OmegaRange As Range, ARRange As Range, TargetAR As Double) As Double
    Dim i As Long
    Dim w1 As Double, w2 As Double
    Dim a1 As Double, a2 As Double

    For i = 2 To ARRange.Rows.Count
        a1 = ARRange.Cells(i - 1, 1).Value
        a2 = ARRange.Cells(i, 1).Value

        If (a1 >= TargetAR And a2 <= TargetAR) Or (a1 <= TargetAR And a2 >= TargetAR) Then
            w1 = OmegaRange.Cells(i - 1, 1).Value
            w2 = OmegaRange.Cells(i, 1).Value

            If a2 = a1 Then
                OmegaAtAR = w1
            Else
                OmegaAtAR = w1 + (TargetAR - a1) * (w2 - w1) / (a2 - a1)
            End If
            Exit Function
        End If
    Next i

    OmegaAtAR = CVErr(xlErrNA)
End Function

Function OmegaAtPhase(OmegaRange As Range, PhaseRange As Range, TargetPhase As Double) As Double
    Dim i As Long
    Dim w1 As Double, w2 As Double
    Dim p1 As Double, p2 As Double

    For i = 2 To PhaseRange.Rows.Count
        p1 = PhaseRange.Cells(i - 1, 1).Value
        p2 = PhaseRange.Cells(i, 1).Value

        If (p1 >= TargetPhase And p2 <= TargetPhase) Or (p1 <= TargetPhase And p2 >= TargetPhase) Then
            w1 = OmegaRange.Cells(i - 1, 1).Value
            w2 = OmegaRange.Cells(i, 1).Value

            If p2 = p1 Then
                OmegaAtPhase = w1
            Else
                OmegaAtPhase = w1 + (TargetPhase - p1) * (w2 - w1) / (p2 - p1)
            End If
            Exit Function
        End If
    Next i

    OmegaAtPhase = CVErr(xlErrNA)
End Function
```

Figure 4: Interpolation VBA code

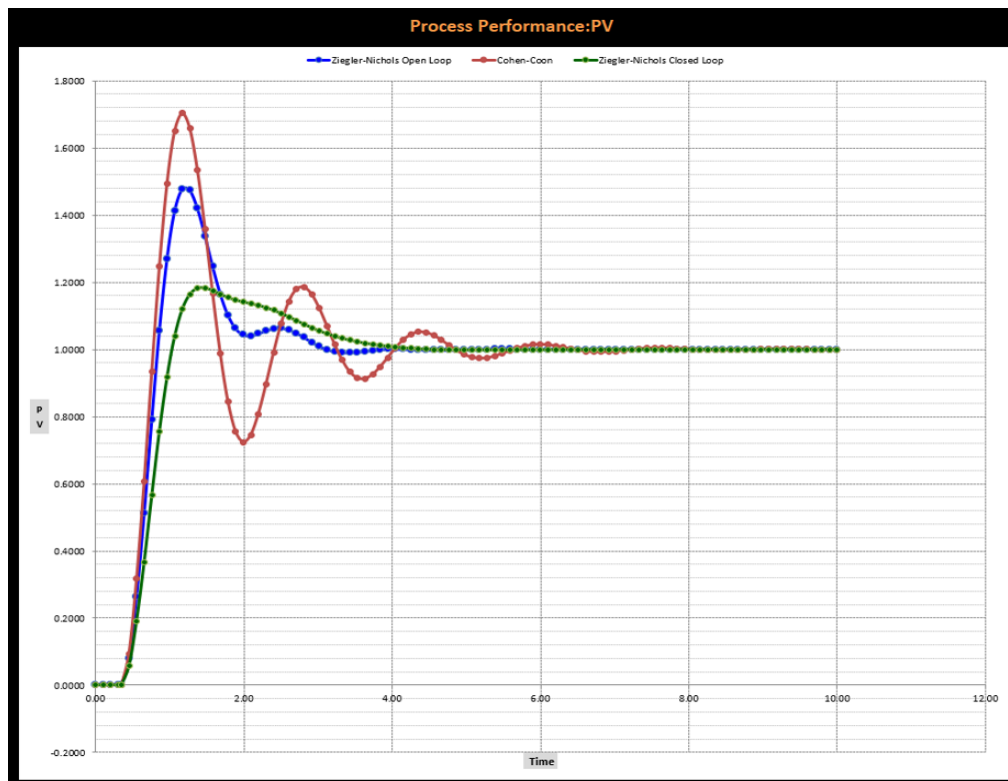


Figure 4: Setpoint change for ZN-Open loop, Cohen-Coon, and ZN-Closed Loop Controllers

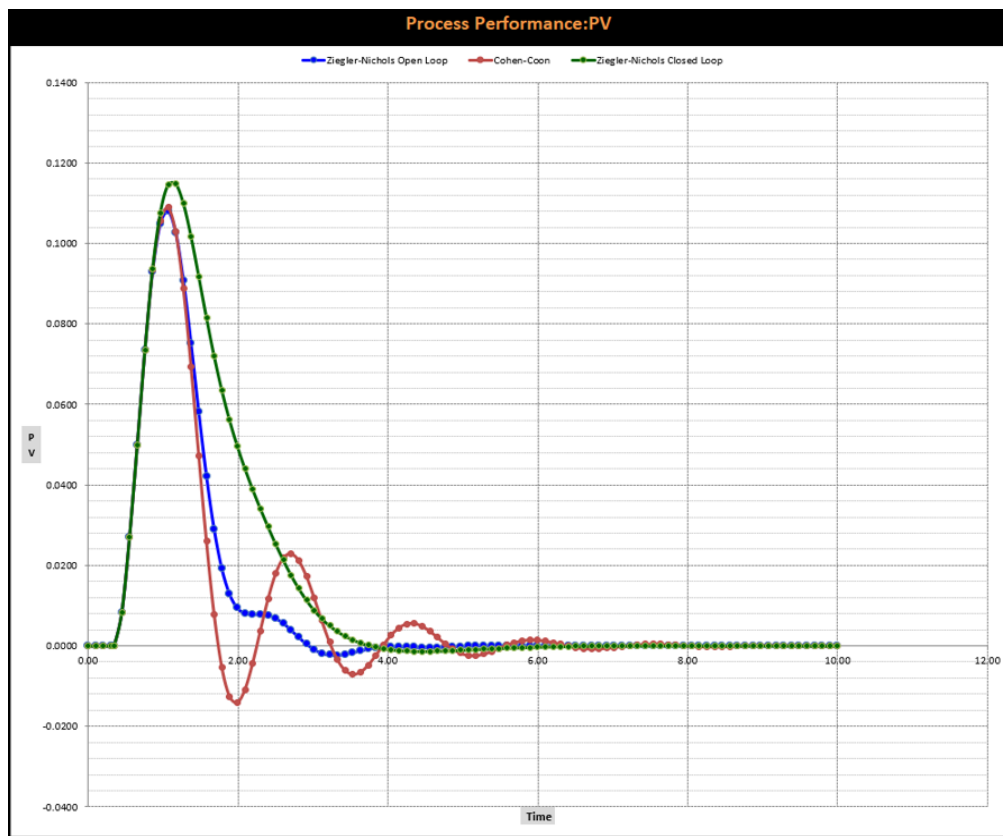


Figure 5: Disturbance change for ZN-Open loop, Cohen-Coon, and ZN-Closed Loop Controllers

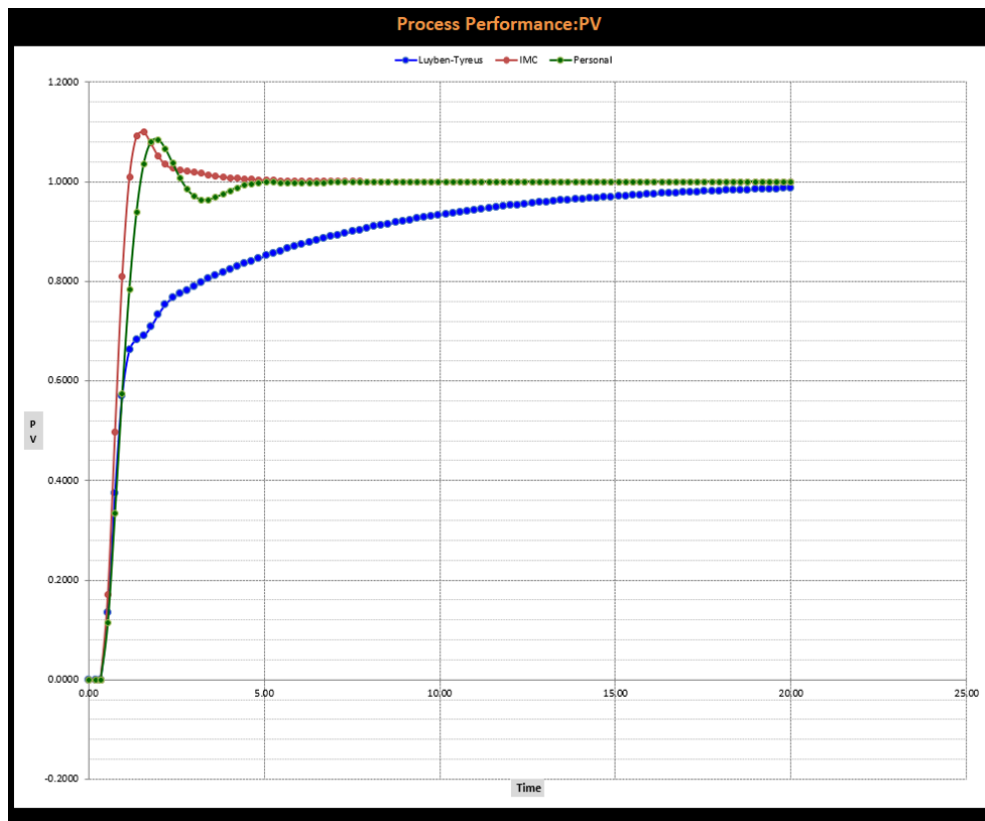


Figure 6: Setpoint change for Luyben-Tyres, IMC, and Personal Controllers

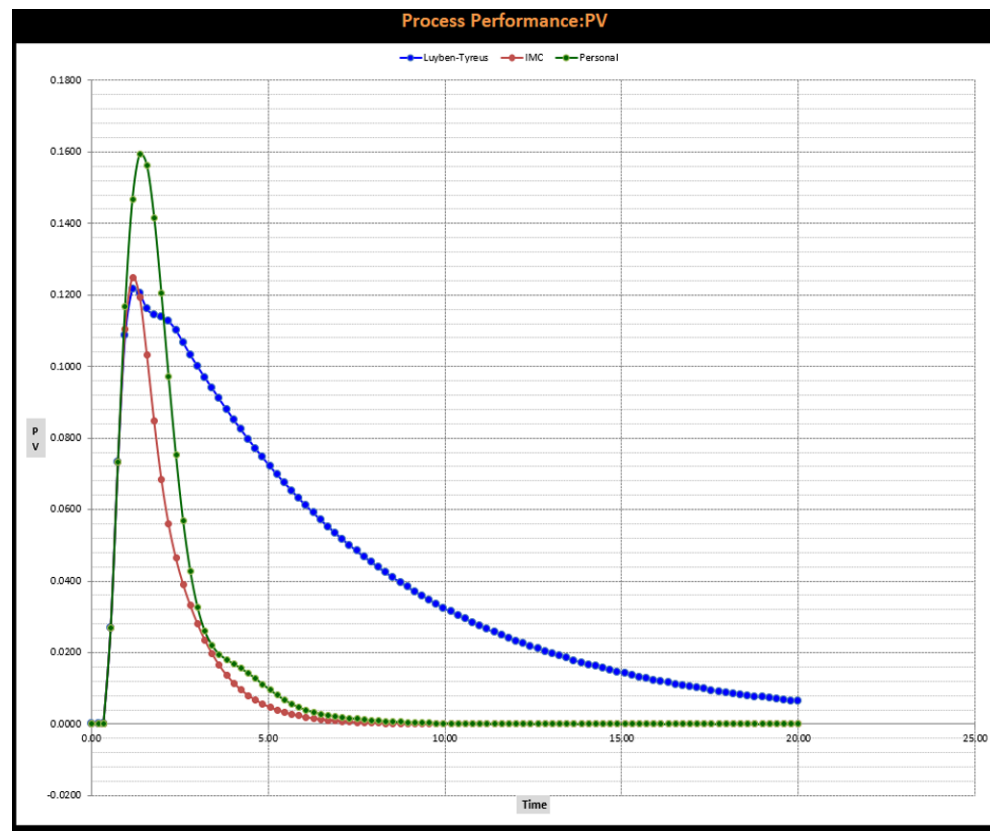


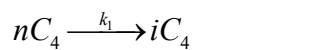
Figure 7: Disturbance change for Luyben-Tyres, IMC, and Personal Controllers

Appendix II: References

- [1] Caracostas, Michael. *CHE Calculator, Version 13*. Excel-based chemical engineering calculation and dynamic modeling tool. University of Illinois Chicago, Department of Chemical Engineering, 2026.
- [2] Caracoitas, Michael. *Introduction to Chemical Process Control*. Chapter 5. University of Illinois Chicago, Department of Chemical Engineering, 2026.
- [3] Caracoitas, Michael. *Introduction to Chemical Process Control*. Chapter 6. University of Illinois Chicago, Department of Chemical Engineering, 2026.
- [4] OpenAI. *ChatGPT* (GPT-based large language model). San Francisco: OpenAI, accessed March 2026. <https://chat.openai.com>.

Problem A: Plant Wide Process Control.

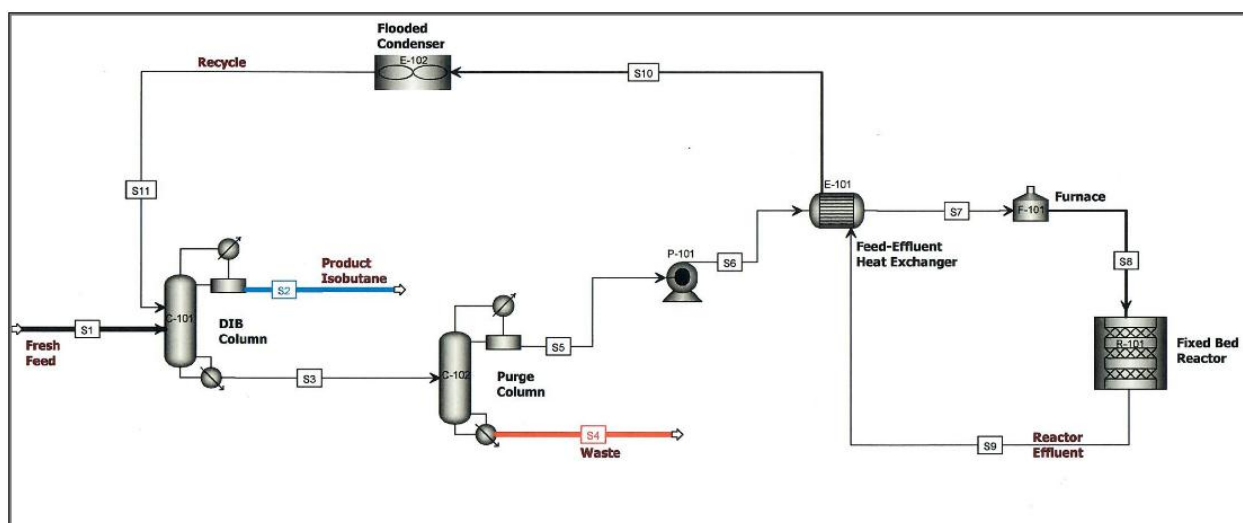
A process flowsheet diagram for the production of iso-butane via the isomerization of n-butane is shown in the Figure below: The isomerization reaction takes place in the fixed bed reactor with pressure drop equal to 15 psi. The kinetics of the reaction are shown below:



$$R_1 = k_{10} \exp\left\{-\frac{E}{RT}\right\} C_{nC_4}$$

$$E = 30,000 \text{ Btu} / \text{lbmol}$$

$$k_{10} = 4 \times 10^8 \text{ l} / \text{hr}$$



To better understand the process, open the attached ASPEN simulation and carefully study the operating conditions of all unit operations. The distillate product from the DeIsoButanizer column (DIB) has a specification of 2 mol% n-butane and is directed to a storage tank. The plant uses cooling water from near-by cooling tower designed with 75% efficiency with hot water inlet temperature of 120 °F and air ambient temperature of 90 °F with relative humidity 70%. A steam plant from a nearby facility supplies saturated steam (with quality 0.95) at three different levels of pressure: low pressure at 50 psig, medium pressure at 150 psig and high pressure at 600 psig. Natural gas from an existing pipe line is used for the fired heater that heats the feed to the reactor. The heat transfer coefficients in all the heat exchangers can be taken as the middle values of the ranges that are listed in the **CHE Calculator**.

Assume that the pressure drop in the heat exchanger is 10 psi both on the tube side and 5 psi on the shell side and the pressure drop in the fired heater is 60 psi. The pressure drop in the liquid line from the pump discharge to the inlet of the fired heater is approximately 8 psi.

This assignment has three parts:

Part A: Flowsheet Modifications

1. First you must modify the simulation to obtain the product specifications and the diameters of the distillation columns. That will help you calculate the height of the liquid in the column sump. Some modifications will also be need to accommodate the additional information on pressure drop given in the problem description above.
2. Second you must create three utilities (cooling water, steam and natural gas) and attach them to the appropriate unit operations. Utility consumption will be used to calculate the steady-state values of the manipulated variables for the process control loops that you will identify in **Part B**.

Part B: Identification and Design of Process Control Loops

Following the procedure that I have outlined in the lectures, calculate the Control Degrees of Freedom for this process and design all process control loops using the additional information outlined below.

Data and Assumptions

1. Both refluxes in the **DIB** column and the **Purge** column are on flow control
2. Residence time in the reflux accumulation drums: 4 **min**
3. Residence time in the distillation column sumps: 2 **min**
4. The waste stream is also on flow control due to environmental restrictions
5. Neither the fresh feed nor the product flow rates are fixed by other plant considerations
6. Heating value of natural gas used in the fired heater: 50.0 **MJ/kg**
7. The heat capacity of water can be taken equal to 4.18 **kJ/(kg C)**
8. Temperature rise in the cooling water is limited to 11 °C

Process Variables Units for Reporting

1. Liquid flow rates must be given in **gpm**
2. Gas flow rates in **lb/hr**
3. Temperature in **deg F**
4. Pressure in **psia**
5. Height in **ft**

For each control loop you have identified write a summary similar to the template shown below:

Template: Design of Process Control Loops

Loop #	Controller Description	Controlled Variable: CV	Set Point Value: SP	Manipulated Variable: MV	Steady-State Normal Value:
1	DIB Column Reflux Flow	Reflux Flow		Reflux Flow	
2					

...

Part C: Process Simulation Calculations

From pilot plant studies it was recommended to increase the reactor feed temperature by 10 °F in order to increase production rate. Change the set point of the temperature controller and answer the following questions after carrying out the simulation with the new reactor feed temperature.

1. Briefly explain your logic in setting up the simulation for the change in the temperature loop set point value from 390 deg F to 400 deg F. Did you have to change the set points of any other controllers? If yes provide a brief explanation as to why and then enter their new values in the template shown below.
2. Run the simulation in order to calculate the new steady-state values of the manipulated values for all controllers and enter the new values in the template shown below

Template: Simulation Calculations after Set Point Change

Loop #	Controller Description	Controlled Variable: CV	Old Set Point: SP	New Set Point: SP	MV	Old Steady State Value	New Steady State Value
1	DIB Column Reflux Flow	Reflux Flow Rate			Reflux Flow Rate		
2							
...							

Solution - Part A: Flowsheet Modifications

The objective of this section was to modify the base Aspen Plus simulation to accurately reflect the physical constraints and equipment dimensions of an industrial isomerization plant. These modifications were necessary to establish a realistic steady-state baseline, ensuring the system has sufficient pressure and inventory to support the process control loops designed in the subsequent stages of the project.

To start, we adjusted the pressure profile of the system. We introduced the pressure drop in the system. In the feed-effluent heat exchanger (E-101), we designated the tube side as the cold side because it carries the reactor feed (S6) that needs to be heated, while the shell side was designated as the hot side as it carries the reactor effluent (S9) providing the thermal energy. We implemented a 10 psi drop on the tube side and a 5 psi drop on the shell side of this exchanger. For the fired heater (F-101), we entered a value of -60 psi into the pressure specification field; in Aspen Plus, entering a negative value for pressure is the standard method to instruct the program to calculate a specific pressure drop rather than a fixed outlet pressure. To account for the 8 psi drop for the liquid discharge line, the P-101 pump discharge pressure was decreased by 8 psi to 657 psia.

With the pressure balance stabilized, we then modified the DIB column (C-101) to hit the required product specification of 2 mol% n-butane in the distillate. Because the changes in system pressure slightly altered the feed conditions, the column was re-optimized by varying the reflux ratio. We set a design specification to fix the n-butane mole fraction at 0.02, which resulted in the reflux ratio adjusting from 7.8 to 8.02. This final converged state provided the baseline flows and duties needed for equipment sizing.

The columns were then sized using the Aspen Process Economic Analyzer (APEA) within the economics tab. This tool maps the simulation results to real-world equipment types, allowing us to obtain accurate column diameters based on tray hydraulics and flooding limits. The sizing analysis provided a bottom section diameter of 8.5 ft for the DIB column and 5.5 ft for the Purge column. Following the simulation run with these dimensions, we extracted the volumetric flow rates of the bottoms streams, S3 and S4, to calculate the necessary liquid inventories.

Finally, we calculated the required liquid sump heights. The sump is the bottom-most part of a column that acts as a reservoir, providing a buffer of liquid to protect downstream pumps and ensure stable reboiler operation. To calculate the height, we used the formula for the volume of a cylinder ($V = \pi \cdot r^2 \cdot h$), rearranged to solve for height. This assumes a standard vertical sump section. The diameter of the bottom section of the column were taken from the column internals, which were auto generated after selecting “Auto Section” when “Design and specify column internals was selected”. By applying the 2-minute residence time requirement, we determined how much liquid height is needed to maintain a safe inventory.

DIB Column (C-101) Sump Height:

- Bottoms Volumetric Flow (S3): 1603.80 ft³/hr
- Column Diameter: 10.25 ft
- Volume Required (2 min): 1603.8 ft³/hr $\times$ (2/60) hr = 53.46 ft³

- Sump Area: $\pi \times (5.125 \text{ ft})^2 = 82.252 \text{ ft}^2$
- Sump Height: $53.46 \text{ ft}^3 / 82.52 \text{ ft}^2 = \mathbf{0.648 \text{ ft}}$

Purge Column (C-102) Sump Height:

- Bottoms Volumetric Flow (S4): $62.72 \text{ ft}^3/\text{hr}$
- Column Diameter: 5.25 ft
- Volume Required (2 min): $62.71 \text{ ft}^3/\text{hr} \times (2/60) \text{ hr} = 2.09 \text{ ft}^3$
- Sump Area: $\pi \times (2.6125 \text{ ft})^2 = 21.44 \text{ ft}^2$
- Sump Height: $2.09 \text{ ft}^3 / 23.76 \text{ ft}^2 = \mathbf{0.0966 \text{ ft}}$

To determine the utility consumption, we start by generating the required utilities in Aspen Plus. The cooling water utility (U-CW) was modeled using an inlet temperature of 81.8°F , which represents the wet-bulb temperature calculated using the "WetBulbTemp" function of the Che Calculator for a 90°F day with 70% humidity. With a maximum allowable rise of 11°C (19.8°F), the outlet temperature was set to 101.6°F . This utility provides the necessary cooling for the flooded condenser (E-102) and the condensers on both the DIB and Purge columns.

The heating requirements for the plant are split between the distillation and reaction sections. The steam utility (U-STEAM) was entered as saturated low-pressure steam at 50 psig to provide the energy for the kettle reboilers on both columns. For the fired heater (F-101), the natural gas utility (U-GAS) was copied from the template for "Fired Heat (1000)", configured with the provided heating value of 50.0 MJ/kg to heat the reactor feed to its target operating temperature of 390°F .

Following the configuration of the site-specific temperatures and physical property assumptions, we extracted the current utility consumption for the entire flowsheet. These results represent the baseline energy requirements of the process in its steady-state converged form, prior to the implementation of any dynamic control loops in Part B.

The cooling water utility (U-CW) shows a total thermal duty of $6.861 \times 10^7 \text{ BTU/hr}$, while the low-pressure steam utility (U-STEAM) required for the reboilers accounts for $6.093 \times 10^7 \text{ BTU/hr}$. Additionally, the natural gas utility (U-GAS) providing heat for the fired heater has a consumption rate of $6.918 \times 10^6 \text{ BTU/hr}$. Converted to gpm and lb/hr, the steady stage utility consumption is 6,925 gpm of cooling water, 66,804 lb/hr of LP steam, and 322 lb/hr of natural gas. Next, the control degrees of freedom will be calculated and control loops will be implemented.

Solution - Part B: Identification and Design of Process Control Loops

The control degrees of freedom were estimated by counting the independent “handles” available in the flowsheet and subtracting items that introduce additional constraints. The process contains 21 mass streams (including internal column streams), 6 energy streams (utilities and heat duties), 10 extra phases (phase changes across column sections and standalone equipment such as the fired heater and flooded condenser), and 2 extra liquid segments (liquid-in/liquid-out sections that do not provide inventory, such as the pump and the liquid-side segment of the heat exchanger). Chapter 2 of *Introduction to Process Control* was used as reference when identifying the degrees of freedom for individual equipment, especially the two columns. After accounting for all the streams, extra phases, and liquid segments, the overall controllable degrees of freedom are:

$$\text{C.D.O.F.} = 21 + 6 - 10 - 2 = 15$$

This means the flowsheet can support up to 15 independent control actions (i.e., control valves or equivalent manipulated inputs) without over-specifying the process. In practice, fewer than the maximum may be implemented because some streams are already committed to specific roles (for example, refluxes are explicitly on flow control), and because certain valve placements are avoided for operability and equipment-protection reasons.

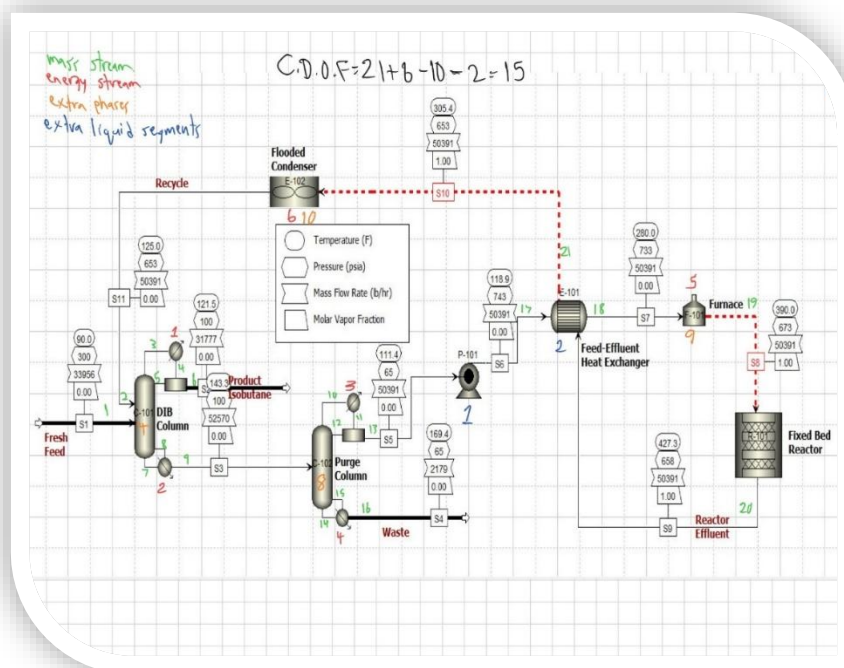


Figure 1: C.D.O.F for production of Isobutane

The control structure is built around the basic requirements of stable operation. Both distillation columns require inventory regulation to prevent dry-out or flooding, so sump levels and reflux accumulator levels must be controlled. Both columns also require pressure control, since column pressure strongly affects relative volatility, condenser behavior, and overall separation. For columns with total condensers, the most direct and standard way to regulate pressure is by manipulating the cooling-water flow to the condenser, because it changes the condensation rate and therefore the vapor holdup/pressure at the top of the column. In addition, the process must meet a product-quality target (isobutane purity in distillate) and satisfy a specified waste-handling constraint, so composition and discharge flow control are included where required.

A key constraint in this flowsheet is that both reflux streams are already on flow control by problem statement, meaning reflux cannot be used as a manipulated variable for level or composition control. That forces the remaining column control objectives to be met using other available handles. For example, steam/reboiler duty remains the primary lever for changing separation strength and composition, but it may also be reassigned to level control in cases where distillate manipulation is impractical (such as when a downstream pump configuration makes throttling a distillate line undesirable). Similarly, because neither the fresh feed nor the product flow is fixed by external plant constraints, these flows can be used as inventory-control handles where appropriate, instead of relying on reflux adjustments.

Downstream of the columns, the reactor section introduces additional variables that must be regulated for safe and consistent operation. The reactor inlet temperature must be maintained to keep the fixed-bed reactor at a stable operating condition, so the natural gas flow to the fired heater is used as the manipulated variable for temperature control. The reactor feed flow is regulated using a valve on the feed line, specifically the stream following the pump due to its the lower volumetric flow compared to the HX effluent, allowing smaller valve sizing. Finally, the recycle/condensation section is used to manage the recycle conditions: the flooded condenser cooling-water flow is used to regulate the recycle temperature, while a recycle-side flow valve provides a practical handle for influencing the overall recycle rate and, in turn, the pressure behavior in the recycle/reactor system. Together, these loops are designed to satisfy the required safety and operating constraints while respecting the limitations imposed by streams that are already committed to flow control. The placement of control valves is shown in the figure below:

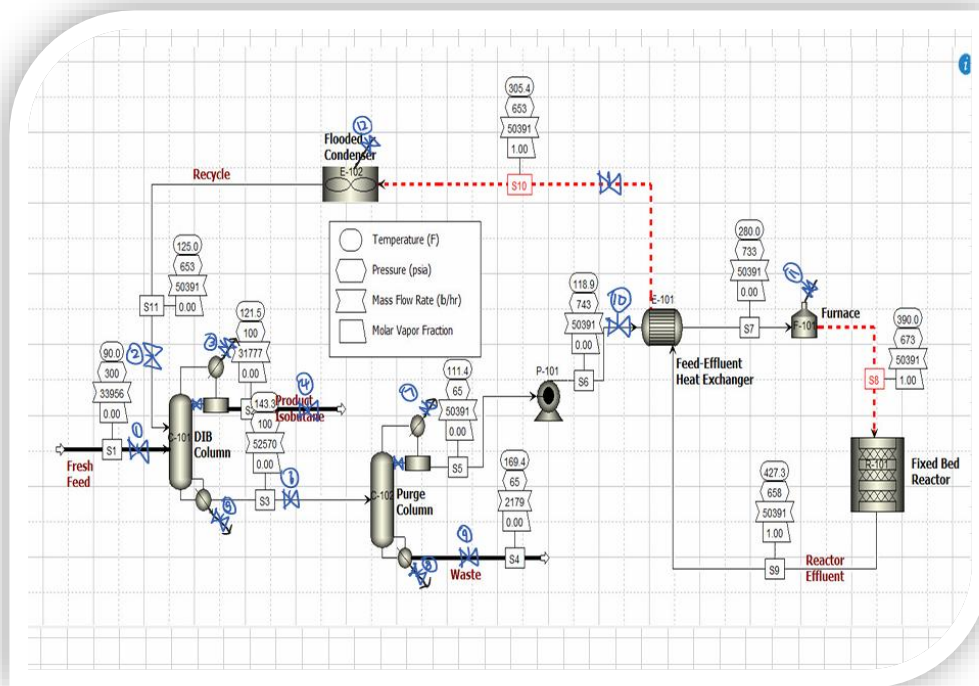


Figure 2: Control Valve placement in plant

Before proceeding with introducing a disturbance, the steady state values must be recorded. For the sake of simplicity, the set points for each manipulated variable were set to their current value. Cooling water flows were imported from Aspen in lb/hr, and converted to gpm using a density of 8.32 lb/gal, which is about the density of water at 90 °F (average of inlet and outlet CW temperature). To get accumulator level, the same procedure as sump level was performed, except the distillate and reflux volumetric flows were added and used to calculate accumulator volume. The table below reports all control loops and steady state values.

Table 1: Control Loop summary

Loop #	Controller Description	Controlled Variable: CV	Set Point Value: SP	Manipulated Variable: MV	Steady-State Normal Value:
1	DIB Column Feed Flow	Sump Level	0.647 ft	Fresh Feed Flow	120.91 gpm
2	DIB Column Recycle Flow	Reactor Pressure	587 psia	Recycle Flow	190.68 gpm
3	DIB Column CW Flow	Column Pressure	100 psia	CW Flow	52,689 gpm
4	DIB Column Distillate Flow	Accumulator Level	14.85 ft	Distillate Flow	122.69 gpm
5	DIB Column Steam Flow	Isobutane Purity	0.02	Steam Flow	56,142 lb/hr

6	Purge Column Feed Flow	Sump Level	0.0966 ft	Feed Flow	199.95 gpm
7	Purge Column CW Flow	Column Pressure	65 psia	CW Flow	14,833 gpm
8	Purge Column Steam Flow	Accumulator Level	9.02 ft	Steam Flow	14,217 lb/hr
9	Purge Column Bottom Flow	Waste Flow	7.82 gpm	Bottoms Flow	7.82 gpm
10	HX Cold Side Feed Flow	Reactor Feed Flow	50,401 lb/hr	HX Cold Side Feed Flow	185.6 gpm
11	Furnace Natural Gas Flow	Reactor Feed Temperature	390 °F	Natural Gas Flow	321.9 lb/hr
12	Flooded Condenser Temperature	Recycle Temperature	125 °F	CW Flow	8,967 gpm

Solution - Part C: Process Simulation Calculations

To account for changes in sump and accumulator levels and control them, calculator blocks were created to estimate the column sump heights and reflux accumulator levels using residence-time holdup calculations. For each column sump, the bottoms volumetric flow rate was multiplied by the specified residence time (converted to hours) to obtain the liquid holdup volume. The sump cross-sectional area was then calculated as $A = \pi(D/2)^2$, using the diameter of the bottom section of each column taken once from the internal column design results and treated as constant. This constant-diameter assumption is reasonable because equipment diameters do not change appreciably for small operating shifts. Accumulator liquid levels were calculated in the same manner, except the holdup volume was based on the sum of the distillate and reflux volumetric flow rates, and the drum diameters were taken once from the Aspen Economic Analyzer results and also assumed constant. The disturbance was introduced by increasing the fired-heater (furnace) outlet temperature setpoint to 400 °F, after which the flowsheet was re-converged with the same set of specifications and Design Specs in place to evaluate the resulting inventory response.

Four external Design Specs (not internal column specs) were used to prevent unsafe inventory deviations. Each Design Spec targeted the pre-disturbance height with a relatively loose tolerance of $\pm 20\%$, which allowed the flowsheet to converge reliably even with several interacting specifications. In most cases, the manipulated variables required only small adjustments; the intent was not tight level control, but rather ensuring that sump and accumulator inventories remained within an acceptable operating range. In the DIB column, sump height was regulated by varying the fresh feed flowrate, since feed flow is not externally fixed and directly changes the rate of liquid accumulation in the bottom section. The DIB reflux drum level was regulated by varying the distillate-to-feed (D/F) ratio, which serves as a proxy for manipulating the distillate draw rate while keeping the reflux flow fixed per the problem statement. In the

Purge column, sump height was regulated using a splitter placed upstream of the column, where the split fraction is the manipulated variable; for the current disturbance, the calculated sump height remained within the tolerance band, so the split fraction did not deviate from its baseline value. The Purge reflux drum level was regulated by varying the reflux ratio, which in Aspen acts as an indirect substitute for reboiler heat input: changing reflux ratio shifts the internal reflux and boilup requirements, making it a practical stand-in for manipulating steam/reboiler duty when a direct bottoms or distillate valve is not used.

Several other control objectives were addressed through Aspen's built-in specifications and utility adjustments. Column pressures were maintained using condenser cooling-water utilities, which Aspen automatically adjusts to meet the specified pressure targets, so no additional Design Specs were required. Product quality in the DIB column was enforced with an internal column specification by manipulating the boilup rate, which is effectively equivalent to manipulating reboiler steam. For the Purge column, the waste bottoms flow was fixed at its pre-disturbance value (treated as the allowable discharge limit) by adjusting the distillate-to-feed ratio, which controls how the feed splits between distillate and bottoms. The fired heater fuel flow and flooded condenser cooling-water flow were also handled automatically by Aspen to meet temperature specifications.

The remaining variables left uncontrolled were the recycle flow and the pre-heat-exchanger flow control. Recycle flow cannot realistically be used to regulate reactor pressure in this steady-state Aspen model because downstream pressure specifications (column pressures and pump pressure constraints) dominate the pressure profile, and no empirical pressure–flow relationship is provided. Similarly, the flow upstream of the exchanger was not explicitly controlled because no recommended setpoint exists and manipulating it strongly would unnecessarily disturb the recycle loop behavior. Finally, the manipulated variables for sump and accumulator heights did not change significantly during this disturbance, largely because the selected tolerances were close to the post-disturbance operating values, meaning the Design Specs acted mainly as safeguards rather than actively driving large changes in the manipulated variables. A table reporting the values of the controlled and manipulated variables after and before the disturbance are shown below.

Table 2: Calculations after Set Point Change

Loop #	Controller Description	Controlled Variable: CV	Old Set Point: SP	New Set Point: SP	MV	Old Steady State Value	New Steady State Value
1	DIB Column Feed Flow	Sump Level	0.647 ft	0.649 ft	Fresh Feed Flow	120.91 gpm	120.91 gpm
2	DIB Column Recycle Flow	Reactor Pressure	587 psia	587 psia	Recycle Flow	190.68 gpm	191.62 gpm
3	DIB Column CW Flow	Column Pressure	100 psia	100 psia	CW Flow	52,689 gpm	41,844 gpm
4	DIB Column Distillate Flow	Accumulator Level	14.85 ft	11.79 ft	Distillate Flow	122.69 gpm	122.69 gpm
5	DIB Column Steam Flow	Isobutane Purity	0.02	0.02	Steam Flow	56,142 lb/hr	44,803 lb/hr
6	Purge Column Feed Flow	Sump Level	0.0966 ft	0.0966 ft	Feed Flow	199.95 gpm	200.2 gpm
7	Purge Column CW Flow	Column Pressure	65 psia	65 psia	CW Flow	14,833 gpm	14,766 gpm
8	Purge Column Steam Flow	Accumulator Level	9.02 ft	9.03 ft	Steam Flow	14,217 lb/hr	14,156 lb/hr
9	Purge Column Bottom Flow	Waste Flow	7.82 gpm	7.82 gpm	Bottoms Flow	7.82 gpm	7.82 gpm
10	HX Cold Side Feed Flow	Reactor Feed Flow	50,401 lb/hr	50,390 lb/hr	HX Cold Side Feed Flow	185.6 gpm	185.8 gpm
11	Furnace Natural Gas Flow	Reactor Feed Temperature	390 °F	400 °F	Natural Gas Flow	321.9 lb/hr	335.2 lb/hr
12	Flooded Condenser Temperature	Recycle Temperature	125 °F	125 °F	CW Flow	8,967 gpm	9,196 gpm

References

- [1] Aspen Technology, Inc. *Aspen Plus User Guide, Version 14*. Bedford, MA: Aspen Technology, Inc., 2023.
- [2] Caracostas, Michael. *CHE Calculator, Version 13*. Excel-based chemical engineering calculation and dynamic modeling tool. University of Illinois Chicago, Department of Chemical Engineering, 2025.
- [3] Caracoitas, Michael. *Introduction to Chemical Process Control*. Chapter 2. University of Illinois Chicago, Department of Chemical Engineering, 2025.
- [4] OpenAI. *ChatGPT* (GPT-based large language model). San Francisco: OpenAI, accessed January 2026. <https://chat.openai.com>.